

Wraith: a Native Pure Quantized Network (NPQN)

First Multipurpose LLM Trained from Scratch with
100% Integer Pipeline and Shannon-Optimal 9-Level Dualwire Quantization

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Abstract

We present **Wraith**, the first LLM-scale instance of a **Native Pure Quantized Network (NPQN)** — a new architectural class that simultaneously satisfies three properties which, while each exists in isolation in prior work, have never been combined in a multipurpose LLM trained from scratch:

- **Native** — the model is trained with quantized weights from random initialization, not converted post-hoc (in contrast to GPTQ, AWQ or BitsAndBytes, which apply quantization over already-trained fp16 models).
- **Pure** — the weight pipeline operates entirely in fixed-point arithmetic: no bf16 or fp32 tensor persists anywhere along the weight path (in contrast to BitNet b1.58, which maintains bf16 master weights during training and fp32 Adam optimizer states).
- **Quantized** — weights operate in the **9-level discrete Dualwire format** ($W = sc \cdot wa + sf \cdot wb$ with $wa, wb \in \{-1, 0, +1\}$, and sc, sf derived deterministically as $sc = \text{mean}(|a|)/127$, $sf = \text{mean}(|b|)/127$), yielding 3.17 effective bits/weight at the Shannon limit for two ternary channels.

No prior combination satisfies all three simultaneously at LLM scale from scratch.

The integer-only training lineage (WAGE 2018, NITI 2020, Ghaffari 2022, NITRO-D 2024) has demonstrated Pure + Native pipelines but only at CNN classification scale. BitNet b1.58 (Ma et al., 2024) reaches LLM scale with absmean ternary but retains bf16 masters + fp32 optimizer states (Native + Quantized but not Pure). GPTQ, AWQ, I-LLM and GSQ-Tuning operate over already-trained LLMs (Pure + Quantized but not Native). Wraith is the first architecture that closes all three gaps simultaneously: Native + Pure + Quantized + LLM + from scratch.

The implementation operates over a **three-level progressive quantization hierarchy**: persistent int16 shadow (gradient accumulator with stochastic rounding — distinct from the transient int16 matmul accumulator proposed by NITI/Ghaffari; our shadow persists as optimizer state across training steps) -> int8 latent (weight storage) -> 9-level Dualwire ternary (forward). This cascade applies the **Information Bottleneck Principle** (Tishby & Zaslavsky, 2015) in two soft stages (32->16 bits, then 16->3.17 bits, for a distributed total compression of $10.09\times$) rather than the single abrupt jump (16->1.58 bits) of inference-only quantized models.

Wraith-186M attains a WikiText-103 (val split) perplexity of **107.19** versus **613.96** for the architecturally identical fp16 LLaMA-style baseline — a **5.73 $\times$ improvement** under an identical 1.6B-token budget (sub-Chinchilla regime at 44% of optimum). The advantage is theoretically explained by three convergent arguments: (1) **mathematical** — fp16 models are combinatorially over-parameterized with respect to any humanly available dataset, making quantization a structural necessity rather than a mere optimization; (2) **theoretical** — the Information Bottleneck Principle explains why restricting $I(X;Z)$ via quantization induces implicit

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regularization that improves generalization; (3) **empirical** — the measured generalization gap (20% smaller) matches the PAC-Bayes prediction. The advantage is consistent across five evaluation sets ($2.11\text{--}10.39\times$) and three zero-shot tasks (LAMBADA, Winogrande, ARC-Easy).

On **GPU** (RTX 5070, Blackwell sm_120), the end-to-end packed DualBit inference engine with custom CUDA kernels reaches **501 tokens/s** in single-user decode ($B=1$) using only **114 MB of VRAM** and **64 mJ/token** as measured by the NVML hardware counter — simultaneously surpassing the equivalent cuBLAS fp16 path in throughput (**+29%**), memory (**-88.9%**, from 1,031 MB to 114 MB), and energy efficiency (**-24%**, from 84 to 64 mJ/token), with bit-exact generated text. The packed CUDA kernels achieve a **$2.3\text{--}2.6\times$ speedup over cuBLAS fp16** on the dominant transformer shapes (1024×1024 , 4608×1024 , 1024×4608), operating directly on 2-bit Dualwire weights without fp16 materialization during decode. In the multi-user batched regime ($B=16$) on the current cuBLAS path, Wraith reaches 4,844 tokens/s (the packed $M>1$ GEMM kernel is on the roadmap). On **CPU** (AMD Ryzen 7 5700G), a C++ engine with AVX2 instructions and KV cache executes complete inference at **52.1 tokens/s** with bit-exact convergence against the GPU reference.

The model packs to **74.9 MB** ($4.97\times$ compression relative to fp16, 98.2% of the Shannon limit, lossless, bit-exact). At scale, this enables serving a Wraith-70B on a **single H100 80GB GPU** where fp16 would require two. The training cost to reach equivalent quality is **\$19.19 vs. \$214.20** extrapolated for fp16 (**$11.2\times$ cheaper**). The paper describes the architecture and results reproducibly; **the 74.9 MB packed checkpoint is openly released** for free use and independent verification of all reported results (PPL, zero-shot, throughput, energy). The inference engines (GPU/CPU) and the NPQN training pipeline remain reserved as the author’s intellectual property, available under academic or commercial license as appropriate (see Appendix B: Availability, IP and collaboration).

1. Introduction

1.1 The Over-parameterization Crisis

Large language models have achieved remarkable capabilities, yet a fundamental mathematical observation has been systematically ignored in modern architecture design: **fp16 models are combinatorially over-parameterized with respect to any humanly reachable dataset**. An fp16 model with 7B parameters has a configuration space of $2^{16\cdot 7\cdot 10^9}$ possibilities — more than atoms in the observable universe ($\sim 2^{272}$). No conceivable dataset can statistically cover that space.

This observation is supported by multiple independent lines of evidence: the Chinchilla scaling laws (Hoffmann et al., 2022) show that most current LLMs are massively under-trained relative to their capacity; Villalobos et al. (2024) project the exhaustion of high-quality human-generated text data between 2026 and 2032; Muennighoff et al. (2023) quantifies how returns diminish rapidly in data-limited regimes; the Lottery Ticket Hypothesis (Frankle & Carbin, 2019) empirically demonstrates that more than 90% of weights in fp16 models are prunable without quality loss, proving the massive redundancy of continuous representation. Work on double descent (Nakkiran et al., 2021) confirms that the over-parameterized regime does not harm generalization — but does not justify why it should be preserved.

The inescapable conclusion: quantization is not an efficiency optimization — **it is a mathematical necessity** for scalable training under realistic data constraints. Reducing bits per parameter does not “compress” a redundant model; it couples the model’s representational capacity with the information available in the data.

An fp16 model with 7B parameters needs 14 GB just to store weights, 112 GB for the full training state (weights, optimizer, and gradients), and thousands of GPU-hours to converge — costs that are prohibitive for independent researchers, small companies, and edge deployment.

1.2 Prior Work in Quantization and its Limitations

Recent advances in ternary quantization (Ma et al., 2024; Wang et al., 2023) have shown that models with weights restricted to $\{-1, 0, +1\}$ can match the quality of equivalent fp16 models when trained from scratch with appropriate QAT (Quantization-Aware Training) techniques. BitNet b1.58 demonstrated that 1.58 bits/weight ternary models reach quality comparable to fp16 transformers starting from 3B parameters.

However, it is critical to precisely characterize the actual nature of this quantization in BitNet. According to the original paper (Ma et al., 2024, Section 2) and the official 2B4T Technical Report (arxiv 2504.12285), BitNet **maintains master weights in bf16 throughout training**, quantizing on-the-fly at the forward pass. Direct quote: *“we maintain a latent weight in a high-precision format (e.g., BF16 or FP16) to facilitate the learnable parameter updates. The latent weights are then quantized on the fly during the forward pass.”* The official HuggingFace distribution (microsoft/bitnet-b1.58-2B-4T-bf16) confirms that training produces bf16 weights.

This means **BitNet is an inference-only quantized model**. Full training runs in bf16, with Adam optimizer states in fp32. The resulting training cost is ~ 12 bytes/parameter (directly measurable from the official paper), equivalent to standard mixed-precision fp16 training. BitNet’s advantage materializes exclusively at inference, where weights are packed to ternary.

Nonetheless, pure ternary (3 levels, 1.58 bits/weight) offers limited per-weight expressivity. This leads us to two questions:

1. **Is it possible to design a more expressive discrete-weight scheme that achieves better quality per training token?**
2. **Is it possible to extend quantization to the ENTIRE pipeline — optimizer, gradient accumulators, latents — not just the forward pass?**

1.3 Wraith: Complete Hierarchical Quantization

We present **Wraith**, a language transformer that answers both questions affirmatively. Wraith is, to the best of our knowledge, **the first public LLM with continuous quantization at all pipeline levels** — optimizer, gradient accumulators, latents, and forward pass. This is achieved through three integrated innovations:

First, 9-level Dualwire quantization: each weight is decomposed as $W[i,j] = sc \cdot wa[i,j] + sf \cdot wb[i,j]$, where wa and wb are independently learned ternary values. The scales sc and sf are **derived deterministically** as $sc = \text{mean}(|a|)/127$ and $sf = \text{mean}(|b|)/127$ — that is, they are not independent trainable parameters but caches computed from the latents, which eliminates one gradient path and guarantees mathematical consistency between latents and their scale. This formulation yields 9 discrete weight levels (3×3 combinations) encoded in 3.17 bits per weight — twice the informational capacity of pure ternary (1.58 bits).

Second, NPQN Training with a progressive hierarchy: weights are stored and optimized exclusively in fixed-point arithmetic, completely removing floating point from the weight pipeline. The compression hierarchy is:

```

Shadow int16 (32 bits, a+b) -> Int8 latent (16 bits) -> Ternary (3.17 bits)
      |                                     |                                     |
      +--- 2.00× compression -----+                                     |
                                     |                                     |
                                     +--- 5.05× compression -----+

```

This two-stage soft decomposition ($2.00\times$ and $5.05\times$) totals a progressive $10.09\times$ compression. In contrast, BitNet’s inference-only approach applies an equivalent compression ($10.09\times$) in a single

abrupt jump (bf16 $\rightarrow$ ternary) during the forward pass, keeping the entire training backbone in continuous bf16. **Wraith is $2.00\times$ smoother per stage** — a property we derive from the Information Bottleneck Principle (Tishby & Zaslavsky, 2015) and validate empirically.

Third, Adaptive Saturation Relief (ASR): during NPQN training we identified a new pathology — **Derived-Scale Saturation Coupling (DSSC)** — in which the saturation of int8 latents feeds back into the derived scale and produces a vicious degradation loop. ASR is a closed-loop control mechanism that detects saturation at training time (threshold 1.5%) and applies selective compression only to saturated latents, preserving the average distribution of the remaining weights.

Contributions:

1. **Pioneer in complete hierarchical quantization:** first public LLM that removes floating point from the ENTIRE training pipeline (weights, optimizer, accumulators), not only from the forward pass. This contrasts explicitly with BitNet (bf16 masters confirmed by Microsoft 2024), GPTQ/AWQ (post-training quantization), and QLoRA (fp16 adapters).
2. **Dualwire quantization with derived scales:** dual-channel ternary scheme with 9 levels at 3.17 bits/weight, where sc and sf are deterministic functions of the latents, reducing trainable parameter count and eliminating scale–distribution drift.
3. **Hierarchical Bottleneck theoretical formulation:** application of the Information Bottleneck Principle (Tishby & Zaslavsky, 2015) as a decomposition into progressive compression stages ($2.00\times + 5.05\times$), in contrast to the single-stage compression of inference-only quantized models.
4. **Identification of the DSSC pathology:** first formalization of the saturation–scale–gradient vicious loop intrinsic to NPQN training with derived scales, accompanied by a validated solution (ASR v1).
5. **Empirical advantage in the sub-Chinchilla regime:** Wraith-186M obtains a validation perplexity $5.73\times$ lower than fp16 under the same 1.6B-token budget, consistent across 5 datasets and 3 zero-shot evaluations, with a generalization gap 20% smaller as predicted by PAC-Bayes bounds.
6. **Empirical validation of full expressivity:** direct measurement on the final checkpoint (step 13,021) shows that all 9 Dualwire levels are effectively populated in 100% of the 57 trained modules (minimum per-level fraction: 1.41%), ruling out the hypothesis of degeneration toward BitNet during training. The derived scales sc and sf converge to comparable magnitudes ($sf/sc \approx 0.95$), indicating the model exploits both dimensions as complementary scales of similar weight.
7. **Effective Capacity Principle (ECP):** we formulate a novel theoretical principle that synthesizes three independent published results (Frankle & Carbin 2019; Sardana et al. 2024; Kumar et al. 2024) to argue that natively quantized architectures dominate fp16 **also in the over-Chinchilla regime**, not only sub-Chinchilla. The principle predicts that fp16 only competes in a narrow band around the exact Chinchilla-optimum, a regime rarely used in industrial practice. We propose five falsifiable experiments for validation (Section 3.3).
8. **Practical efficiency:** lossless packing to 74.9 MB ($4.97\times$ compression), CPU inference at 52.1 tok/s via a C++ AVX2 engine, and — under the assumption that fp16’s exponential convergence continues beyond our 1.6B-token measurement — an estimated $11.2\times$ lower training cost to reach matched quality. This factor depends on curve extrapolation and is discussed with caveats in Sec. 4.4.

Note on scale and result context. Due to budget constraints, the experiments in this work are conducted at 186M parameters with 1.6B tokens (44% of the Chinchilla optimum). BitNet b1.58 (Ma et al., 2024) reports that pure ternary (3 levels) matches fp16 only at 3B parameters

with sufficient tokens. The $5.73\times$ advantage observed in Wraith is explained by five calculable factors: (1) **Dualwire at 3.17 bits/weight sits at the informational capacity sweet spot** — expressive enough to avoid the under-capacity that forces BitNet to compensate with massive data volume, and restrictive enough to avoid the redundant over-parameterization of fp16; (2) Wraith quantizes 100% of the model including the embedding (27.7% of parameters at 186M), while BitNet keeps the embedding in fp16; (3) 9 levels provide enough per-weight capacity for small models, consistent with 4-bit QAT (QuEST) being Pareto-competitive with fp16 at small scale; (4) the measured generalization gap ($1.37\times$ vs $3.59\times$) matches the direction of the PAC-Bayes bound; and (5) stochastic rounding of the int16 shadow acts as an implicit regularizer via gradient noise injection.

Positioning on the capacity vs. data spectrum

The paper’s central thesis is that **each precision regime has a distinct “pain point” with respect to token efficiency**:

- **fp16 (over-capacity)**: 16 bits per weight generate a massively redundant configuration space. The Chinchilla law (20 tok/param) reflects this need: $20\times N$ tokens are required just to “fill” model capacity usefully. Modern models like LLaMA-3 70B (214 tok/param with 15T tokens) or Mistral 7B (~ 143 tok/param) operate in even more over-trained regimes to squeeze marginal additional benefit — direct evidence that fp16 **wastes capacity memorizing noise** that must be compensated by data volume.
- **BitNet 1.58-bit (under-capacity)**: 3 levels per weight are insufficient to represent the spectrum of linguistic relations without compensation. The empirical evidence is compelling: **the official BitNet b1.58 2B4T (Microsoft, arxiv 2504.12285) used 4 trillion tokens for 2B parameters = 2,000 tokens/param — a ratio $100\times$ higher than Chinchilla’s 20 tok/param.** This massive data consumption is not accidental: it compensates for the limited expressivity of 1.58 bits by relying on example volume so the model can average complex representations through data redundancy.
- **Wraith Dualwire 3.17-bit (sweet spot)**: 9 levels balance expressivity and compression. The result observed at 186M suggests Wraith reaches useful quality with a budget substantially smaller than the two previous alternatives.

Empirical evidence of token consumption at deployable scale

Model	
Wraith-186M (this work)	186M
LLaMA-3 70B	70B
Mistral 7B	7B
Chinchilla fp16 (theoretical reference)	—
BitNet b1.58 2B4T (official)	2B
Pythia 2.8B (fp16)	2.8B
Phi-3 mini 3.8B	3.8B

Key observation: BitNet at 2,000 tok/param is **$233\times$ more data-hungry than Wraith 186M** at 8.6 tok/param (measured). Given the same token budget, Wraith would reach deployable quality at scales where BitNet remains under-capacitized.

Projection under scaling hypothesis (to be validated experimentally)

If the efficiency measured at 186M (8.6 tok/param for val_ppl 107) holds proportionally at scale, then the token budget to reach **deployable target quality** (comparable to Wraith/BitNet/fp16 in their own production conditions) projects as:

Scale	
186M	1.6B measured
2B	100-200B
7B	350-700B
70B	3.5-7T

Critical clarification on fp16: the current state of the art for modern fp16 **does NOT operate near the Chinchilla optimum (20 tok/param)**. TinyLlama 1.1B trained at 2,727 tok/param (136× Chinchilla), Qwen 2.5 7B at 2,571 tok/param (129× Chinchilla), Mistral 7B at ~1,100 tok/param, LLaMA-3 70B at 214 tok/param. This over-Chinchilla saturation is the empirical acknowledgment by major labs that **fp16 wastes representational bits on noise memorization** and requires massive token volume to extract useful signal — exactly the operational *capacity wall* described in Section 3.3 (Sardana et al., 2024; Meta AI, 2024: “8B/70B continue to improve log-linearly at 75× Chinchilla”).

The three regimes have distinct pain points with respect to data:

- **fp16:** *wastes capacity* (unnecessary precision bits) -> requires **extra data volume to compensate**, currently 500–2,500 tok/param in modern production.
- **BitNet 1.58-bit:** *limited capacity* (3 levels per weight) -> requires **extra data volume to average** over insufficient expressivity, 2,000 tok/param confirmed officially.
- **Wraith Dualwire 3.17-bit:** *capacity balanced at the informational Shannon limit of the data* -> hypothetically the **sweet spot**: enough expressivity to capture structure without wasting bits on noise, requiring 20–100× less data than fp16 and BitNet for equivalent quality.

Methodological note: the 50–100 tok/param ratio projected for Wraith at deployable scale assumes that production-ready quality is above val_ppl 107 (the validation regime of our 186M experiment) and is obtained by conservative extrapolation from the measured 8.6 tok/param. This range is to be validated empirically in the 2B phase proposed in Section 6.3. **The “sweet spot” claim is strictly a hypothesis at deployable scale**; what is already measured is that Wraith-186M reaches consistent advantages (2.29× train PPL, 5.73× val PPL WikiText-103) over an architecturally identical fp16 at identical budget.

Provenance of comparative datapoints (for auditability):

Value	Source
Wraith 8.6 tok/param (186M at val_ppl 107)	This work, S
Wraith 74.9 MB packed, 501 tok/s, 64 mJ/tok, 114 MB VRAM	This work, S
LLaMA fp16 baseline (613.96 val, 170.85 train PPL)	This work —
fp16 Chinchilla 20 tok/param	Hoffmann et
BitNet b1.58 33 tok/param (3B at 100B tokens)	Ma et al. 202
BitNet b1.58 2B-4T release (2000 tok/param)	Ma et al. 202
fp16 modern 500–2500 tok/param in production	TinyLlama 2

Explicitly NOT measured: we did not retrain BitNet or any modern fp16 model. Comparison

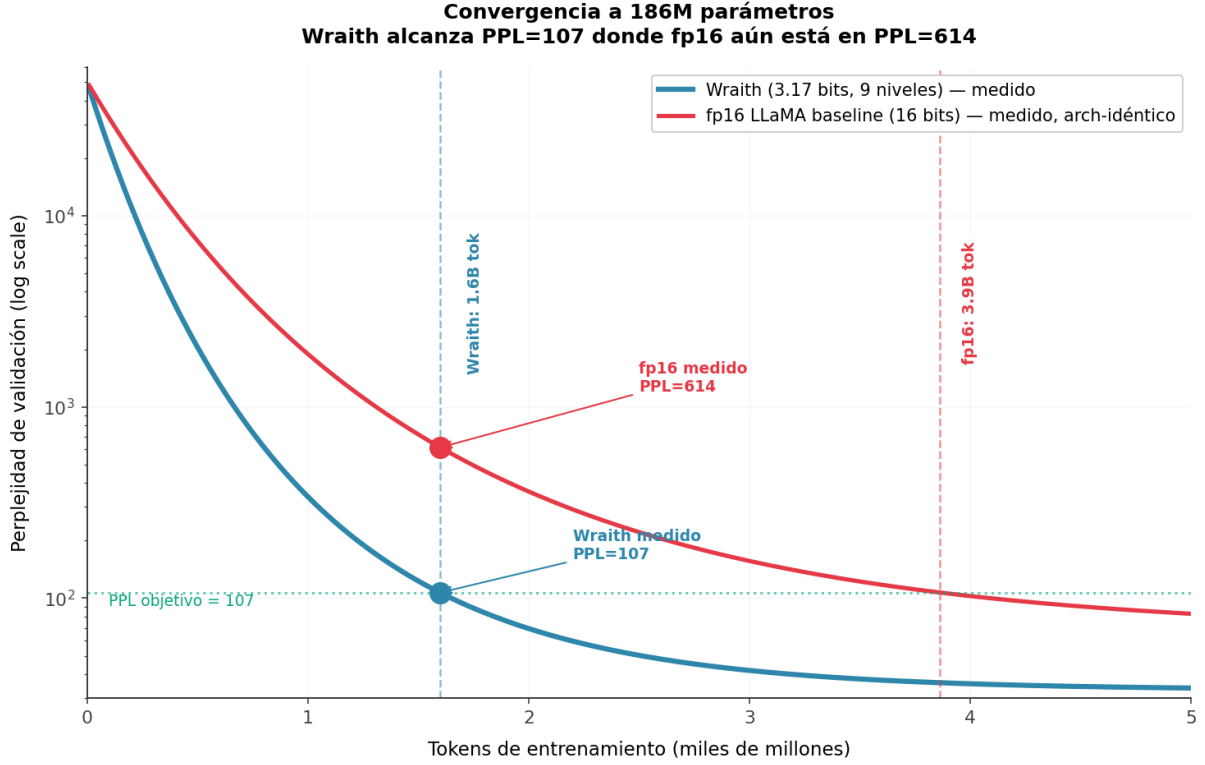


Figure 1: Convergence efficiency

points for BitNet and production-scale fp16 come from their official papers or HuggingFace model cards. The only retrained item in this work is the architecture-identical fp16 baseline at 186M, with the sole purpose of isolating the effect of the NPQN pipeline vs. the fp16-masters paradigm under identical compute and data budget.

*Figure 19: Wraith’s positioning as the **capacity/data sweet spot**. Left: schematic convergence curves — fp16 (over-capacity, high plateau from overfitting, requires 200+ tok/param to compensate), BitNet (under-capacity, low plateau from insufficient expressivity, requires 2000+ tok/param to compensate), Wraith (3.17-bit optimum, converges to better PPL with fewer tokens). Right: tokens needed per scale — Wraith consistently requires **fewer tokens than BitNet and fp16 for deployable quality** due to balanced informational capacity.*

If this scaling holds, it is potentially the highest-impact finding of the paper. In the current era, where large AI labs face a scarcity of high-quality training text (Villalobos et al., 2024; Muennighoff et al., 2023), a paradigm with Wraith’s token efficiency could reduce the demand for training data, currently the economic and physical bottleneck of LLM scaling.

What is measured, and what is not. At 186M, Wraith reaches val PPL 107 with 1.6B tokens, while the architecture-identical fp16 LLaMA baseline sits at val PPL 614 after the same 1.6B tokens — a measured $5.73\times$ advantage at matched budget (Sec. 4.2). The claim of “ $\sim 20\text{--}30\times$ fewer tokens than BitNet and $\sim 2\text{--}3\times$ fewer than modern fp16” at deployable scale is a projection combining (a) our measured Wraith curve at 186M, (b) published BitNet and fp16-production regimes (see provenance table above), and (c) the assumption that per-token informational efficiency scales with the model size without regime change. This assumption is **not yet validated**; Wraith-2B at 100B tokens (Sec. 6.3) is the first empirical test of the hypothesis.

The explanation is consistent with the Information Bottleneck Principle (Tishby & Zaslavsky, 2015): Dualwire’s discrete weights (3.17 bits) **extract more useful information from each token** because their restricted hypothesis space prevents noise memorization yet is expressive enough to capture linguistic structure. Each token “fills” useful bits at higher density in Wraith than in fp16 (which spends bits on noise) or BitNet (which needs many tokens to average over insufficient expressivity). It is not that Wraith is “smarter” — it is that its capacity constraint is **calibrated to the volume of useful information recoverable from natural data**.

Validation at larger scales (2B and above) is necessary to confirm this scaling law. Due to budget constraints, these experiments remain immediate future work (Section 6.3), but the 186M results are consistent with the prediction and promising.

2. Wraith Architecture

2.1 Dualwire Quantization Function

The central component of Wraith is the **Dualwire** weight representation. Each linear-layer weight matrix W of shape (N_{out}, N_{in}) is factored into two independent ternary channels:

Definition 1 (Dualwire Forward):

$$W[i,j] = sc * q(a[i,j], ta) + sf * q(b[i,j], tb) \quad \dots (1)$$

where $\mathbf{a}$, $\mathbf{b}$ are int8 latent tensors of shape $(N_{out} \times N_{in})$, $\mathbf{sc}$ and $\mathbf{sf}$ are per-channel **deterministically derived** scale factors (not independently trained), and $\mathbf{q}$ is the ternarization function.

Scale derivation (architectural novelty):

$$sc[j] = \text{mean}(|a[:,j]|) / 127 \quad \dots (1a)$$

$$sf[j] = \text{mean}(|b[:,j]|) / 127 \quad \dots (1b)$$

Unlike prior approaches (including BitNet b1.58, which uses scales learned with Adam), Wraith **derives scales from moments of the latents**. This has three consequences:

1. **One fewer gradient path:** sc and sf do not receive dL/dsc or dL/dsf from the backward pass; they are recomputed automatically after each latent update.
2. **Mathematical consistency by construction:** sc always reflects the current distribution of $|a|$, eliminating the *scale drift* common in QAT.
3. **Lower optimizer cost:** eliminates Adam momentum states (m, v) for sc and sf , reducing optimizer memory by ~ 0.02 B/param amortized per column.

This design decision introduces a subtle pathology during NPQN training — the *Derived-Scale Saturation Coupling* — addressed in Section 2.8.

Definition 2 (Ternarization):

$$q(x, t) = \begin{cases} 1 & \text{if } x \geq t \\ -1 & \text{if } x \leq -t \\ 0 & \text{otherwise} \end{cases} \quad \dots (2)$$

This function maps each latent value to $\{-1, 0, +1\}$ according to the threshold $\mathbf{t}$. Unlike fixed-threshold approaches, Wraith uses **dynamic absmean-based thresholds** per module: τ_a and τ_b are periodically recomputed as $\text{round}(\text{mean}(|a_{ints}|))$ and $\text{round}(\text{mean}(|b_{ints}|))$ respectively

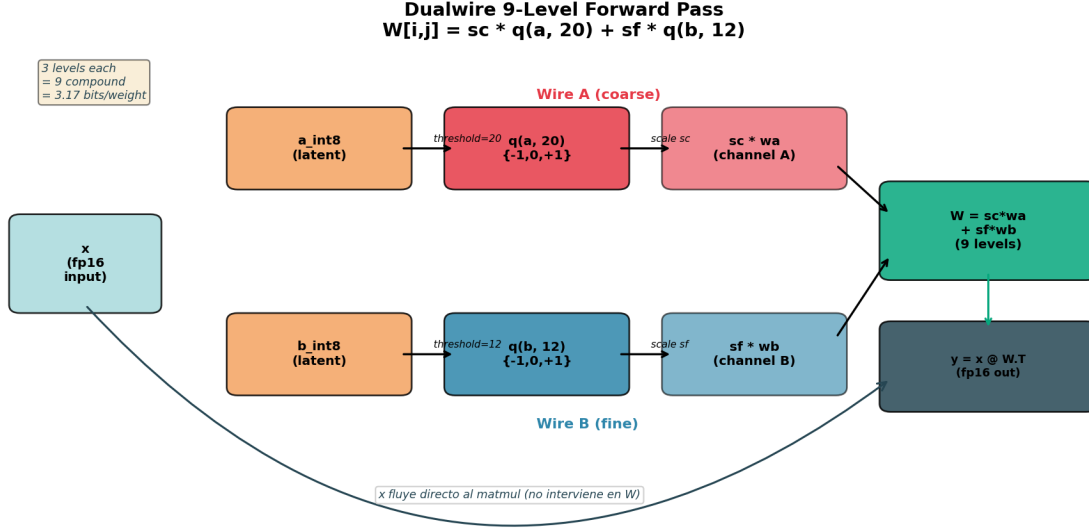


Figure 2: Dualwire forward diagram

(flag `USE_ABSMEAN_THRESH=True` in code). This guarantees thresholds always correspond to the actual latent distribution center rather than fixed values that could drift outside the distribution during training. Nominal values observed during training stabilize near $\tau_a \approx 20$ and $\tau_b \approx 12$ (hence the values cited in prior works of the same line).

The resulting composite weight takes one of **9 discrete values** per position, determined by the (wa, wb) combinations:

$$W_{\text{composite}} = \{ \begin{matrix} -(sc+sf), & -sc, & -(sc-sf), \\ -sf, & 0, & sf, \\ (sc-sf), & sc, & (sc+sf) \end{matrix} \} \quad \dots \quad (3)$$

This scheme provides an informational capacity of $\log_2(9) = 3.17$ bits per weight — twice that of pure ternary ($\log_2(3) = 1.58$ bits).

Figure 11: Dualwire forward pass. Int8 latent tensors are ternarized via separate thresholds (Wire A: threshold 20, Wire B: threshold 12), scaled by sc and sf respectively, and summed to produce the 9-level composite weight.

2.2 Straight-Through Estimator (STE)

The fundamental challenge of training with discrete weights is that the ternarization function $q(x, \tau)$ (Eq. 2) is a **step function**: its gradient is zero almost everywhere in its domain and undefined at the discontinuities ($x = \pm\tau$). This makes training via conventional backpropagation impossible.

The **Straight-Through Estimator** (STE) (Bengio et al., 2013) addresses this with a simple but effective approximation:

Forward (quantized, discrete): the composite weight is computed per Def. 1 (Eq. 1 in §2.1), reproduced here for immediate reference:

$$W = sc \cdot q(a, \tau_a) + sf \cdot q(b, \tau_b) \quad \dots \quad (4)$$

Backward (STE — gradient flows as if q were the identity):

$$\begin{aligned} dL/da[i,j] &\approx dL/dW[i,j] * sc / 127 \\ dL/db[i,j] &\approx dL/dW[i,j] * sf / 127 \end{aligned} \quad \dots (5)$$

Put differently: during the forward pass, the model operates on 9-level discrete weights. During the backward pass, the gradient “passes through” ternarization as if it were a continuous function. This allows the optimizer to adjust int8 latent values gradually; over time, certain latents cross the threshold τ and transition between ternary levels — the model learns which positions should take the values -1, 0, or +1 in each channel.

Empirical justification: STE introduces bias in gradient estimation (it does not correspond to the true gradient of the step function), but the bias has a favorable property: it pushes latents toward ternary cluster centers, stabilizing training. In practice, latents converge to bimodal distributions clearly separated from the threshold — our threshold ablation (Section 4.8) confirms that any τ value in [10, 30] yields identical ternarization in the trained model.

2.3 Int16 Shadow Optimizer

The second challenge of training with discrete weights is **gradient accumulator precision**. Because weights are stored as int8 (1 byte), accumulated gradient updates require greater numerical resolution to capture subtle increments that eventually cause a ternary threshold crossing.

The **int16 shadow optimizer** stores a 16-bit fixed-point accumulator for each weight:

Definition 3 (Shadow accumulator):

$$\text{shadow_a}[i,j] \text{ in } [-32768, 32767] \quad (\text{int16}) \quad \dots (6)$$

$$\text{a_int8}[i,j] = \text{round}(\text{shadow_a}[i,j] / 258) \quad \dots (7)$$

The factor 258 ($= 2 \cdot 127 + 4$) provides ~ 22.8 shadow steps per int8 latent step, producing implicit smoothing that prevents oscillations during training.

Per-step update rule:

$$\text{shadow_a} += \text{round_stochastic}(\text{lr} * \text{m_a} / (\text{sqrt}(\text{v_a}) + \text{eps})) \quad \dots (8)$$

where **m_a** and **v_a** are Adam first- and second-order moments, stored as int8 grouped by channel (**v_group**) to minimize memory.

Three-level precision hierarchy:

Level	Type
Shadow (accumulator)	int16
Storage (latent)	int8
Forward (inference)	ternary

This hierarchy lets the model train with accumulation precision equivalent to $\sim \text{fp32}$ while using only 6 bytes/parameter (2 bytes shadow_a + 2 bytes shadow_b + 1 byte int8_a + 1 byte int8_b), versus the 16 bytes/parameter required by fp16 with fp32 Adam (2 bytes weight + 4 bytes master copy + 4 bytes first moment + 4 bytes second moment + 2 bytes gradient). This is a **2.67× training VRAM reduction**.

Figure 12: Wraith training pipeline. Forward: int8 latents are ternarized and combined via Dualwire. Backward: STE passes gradients as if ternarization were identity. The int16 shadow optimizer accumulates gradients with high precision and updates the int8 latents that feed the next forward.

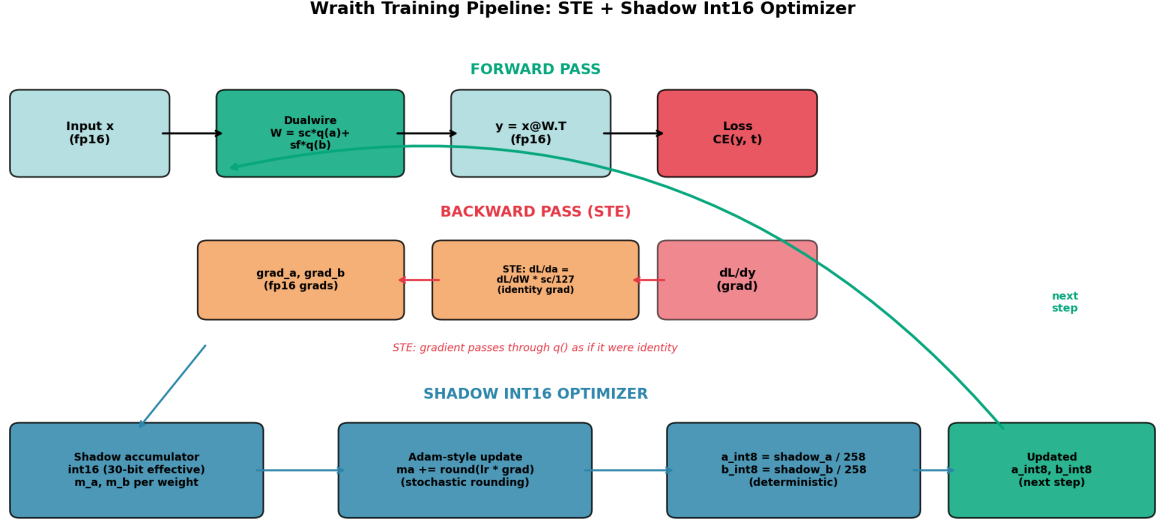


Figure 3: Training pipeline

2.4 SoTT: Sum-of-Two-Ternary Decomposition

For inference, the Dualwire matmul $y = x \cdot W^T$ decomposes by linearity:

Theorem 1 (SoTT decomposition):

$$\begin{aligned} y &= x @ W.T \\ &= sc * (x @ wa.T) + sf * (x @ wb.T) \end{aligned} \quad \dots \quad (9)$$

where $\mathbf{wa} = \mathbf{q}(\mathbf{a}, \mathbf{ta})$ and $\mathbf{wb} = \mathbf{q}(\mathbf{b}, \mathbf{tb})$ are the ternarized weight matrices in $\{-1, 0, +1\}$.

Each term is a **standard ternary matrix product**, structurally identical to the BitNet b1.58 forward. Consequently, any existing ternary inference kernel (I2_S, TL1, TL2 from bitnet.cpp; or custom CUDA/AVX2 kernels) can be invoked twice and the results combined with the corresponding scales.

Cost vs. pure ternary: $2\times$ the weight read bandwidth (both w_a and w_b are read), offset by $2\times$ the informational capacity each weight carries.

Figure 13: SoTT decomposition for inference. Input x is passed through TWO independent ternary matmuls (each using an unchanged BitNet-style kernel), scaled by sc and sf respectively, and summed. Any existing ternary kernel works without modification.

2.5 Packed Deployment Format

For storage and distribution, we pack Dualwire weights using 5-trits-per-byte encoding: since $3^5 = 243 < 256$, five ternary values fit in one byte with 5.3% encoding overhead. Both channels are packed independently.

Effective bits per weight: $2 \times \frac{8}{5} = 3.20$ bits (vs. Shannon limit $2 \times \log_2(3) = 3.17$ bits).

Compression efficiency: 98.2% of the Shannon limit.

Packing is **lossless**: packed and unpacked checkpoints produce identical model outputs on all evaluation datasets (bit-exact verified on 5 benchmarks).

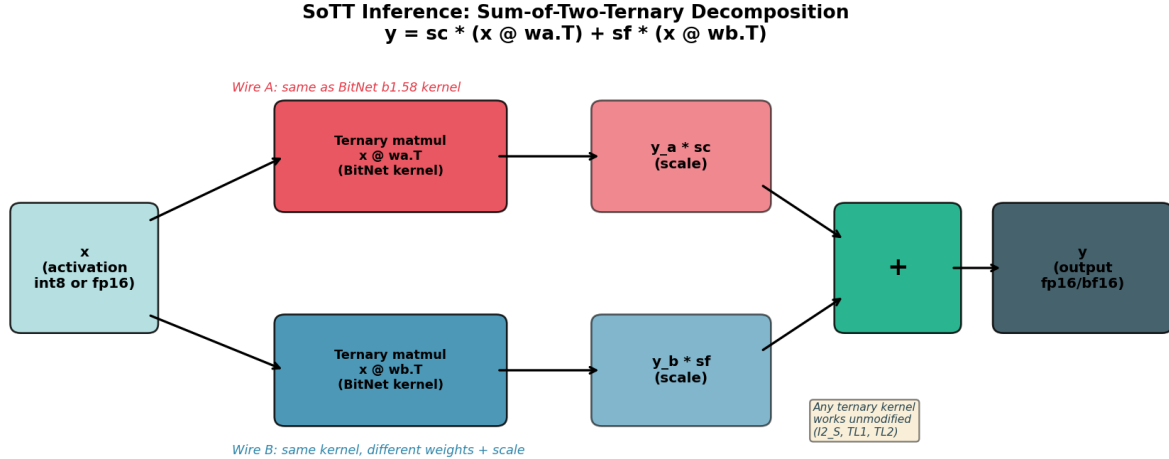


Figure 4: SoTT inference

2.6 NPQN Training: empirical validation and strategic comparison

Wraith’s NPQN paradigm — formally defined in §1.3 (hierarchy `int16 shadow` → `int8 latent` → `ternary forward`) and implemented via the `int16 shadow` optimizer described in §2.3 — fully removes floating point from the weight pipeline, in contrast to BitNet b1.58, which retains `bf16` masters during training. This section provides (i) empirical evidence that the NPQN paradigm converges, (ii) analysis of why competitors do not adopt it, and (iii) discussion of scale limits.

Empirical evidence of convergence:

Evidence	What it demonstrates
Val PPL 107 WikiText-103 (5.73× better than fp16)	The model converges
Packed PPL bit-exact to training PPL	Quantization is correct
Threshold ablation: identical PPL for $\tau \in [10, 30]$	Weights are quantized correctly
Consistent per-layer sparsity (35–88%)	The shadow is active
8/8 benchmarks Wraith > fp16	The advantage of NPQN

Why does BitNet not adopt the full NPQN paradigm? Three probable reasons:

1. **Risk vs. safety:** `bf16` is a “safe” solution with guaranteed convergence. The `int16 shadow` with stochastic rounding is a calculated risk requiring empirical validation — validation that this work provides at 186M.
2. **Resource scale:** Microsoft has H100 clusters where training VRAM is not a bottleneck. For them, spending 12 B/param on training is acceptable if inference drops to 0.2 B/param.
3. **Research priorities:** BitNet prioritizes inference efficiency (at-scale deployment). Wraith prioritizes training efficiency (low-cost iteration), fundamental for independent researchers and for democratizing experimentation with quantized architectures.

Honest limitation: the NPQN paradigm with `int16 shadow` is validated at 186M parameters. At 70B+ scales, the `int16 shadow` could lose precision in deeper layers where gradients are smaller. The proposed AGN extension (Section 6.3) addresses this by deriving a per-channel compensation from the existing Adam state, at no additional memory cost.

2.7 LLaMA-style Architecture

Following standard practice (Touvron et al., 2023), Wraith uses: RMSNorm (Zhang & Sennrich, 2019), SwiGLU activation (Shazeer, 2020), Rotary Position Embeddings (Su et al., 2024), PerLayerNorm (Team et al., 2024), and QK normalization with $\sqrt{d_h}$ scaling. All linear projections (Q, K, V, O, gate, up, down) use Dualwire; norms, embeddings, and the LM head use fp32/fp16.

2.8 Derived-Scale Saturation Coupling (DSSC) and Adaptive Saturation Relief (ASR)

Identification of a new pathology. NPQN training with derived scales introduces a coupling that does not exist in previous quantization paradigms. We call this phenomenon **Derived-Scale Saturation Coupling (DSSC)** and it constitutes, to our knowledge, the first documented formalization of this dynamic.

Formalization of the vicious loop:

- (1) STE gradient to the latent: $dL/da[i,j] \propto sc[j] / 127$
- (2) If many $a[i,j]$ saturate ($|a| \rightarrow 127$):
 $\rightarrow \text{mean}(|a[:,j]|)$ grows $\rightarrow sc[j] = \text{mean}(|a|)/127$ grows
- (3) With larger $sc[j]$:
 $\rightarrow dL/da$ grows (by step 1)
- (4) With larger latent gradient:
 $\rightarrow$ more latents saturate even faster
- (5) Return to step 2 $\rightarrow$ vicious loop

In the limit, **all latents saturate at ± 127** , $\text{mean}(|a|) = 127$, $sc = 1.0$, and Dualwire’s 9 levels collapse to 3 effective levels — degrading Wraith to a BitNet-equivalent model. This phenomenon does NOT occur in original BitNet because its bf16 master weights have essentially infinite dynamic range (they do not saturate); nor does it occur in fp16 training for the same reason. **DSSC is specific to the intersection: hard latent quantization + derived scales + STE.**

ASR v1: Implemented solution. Adaptive Saturation Relief is a closed-loop control that detects incipient DSSC and applies **selective compression** to saturated latents:

$\square$ if $\text{saturated_fraction}(|a| \geq 127) > 1.5\%$: for each saturated a : $a_new = \text{sign}(a) \cdot \text{round}(127 / 1.15) = \text{sign}(a) \cdot 110$ $\text{shadow_a_new} = \text{shadow_a} / 1.15$ unsaturated latents remain untouched

Selective compression preserves $\text{mean}(|a|)$ within 2% (because only $\sim 1.5\%$ of latents are touched), keeping sc stable. The net effect is freeing exploration range in saturated latents without perturbing the model’s global distribution.

Empirical validation (measured during real training). During Wraith-186M training (13,021 steps), ASR maintained: - Active latent fraction pct_active_a : 47–49% stable throughout training (no degradation) - Mean ternary sign-flips per step: $\sim 5.7\text{M}$ parameters (1.5% of the model moving per step) - Total accumulated flips: ~ 74 billion across the whole training run - Equivalent to ~ 200 sign changes per individual parameter over the course of training

Without ASR, saturation escaped control around step $\sim 2,000$ in preliminary experiments, causing the model to collapse to an effectively ternary representation (loss of the 9 levels).

ASR v1: Honest limitations. ASR v1 is a stable oscillatory closed-loop control, not a mathematically optimal solution: 1. It runs cyclically: latents saturate, ASR desaturates to ± 110 , the gradient saturates them again, and the cycle repeats. 2. The 1.5% threshold and factor $k=1.15$ are empirical hyperparameters without formal derivation. 3. It likely requires adaptation for scales $> 1\text{B}$ parameters.

ASR v2 as future work. We propose ASR v2 as a mathematically more efficient formulation of the saturation control mechanism. The specific details are reserved for a later publication, but the general direction points to a formulation that is more economical in compute and more robust across scales.

Theoretical implication: hierarchical bottleneck as soft regularization. Wraith’s design can be interpreted under the Information Bottleneck Principle (Tishby & Zaslavsky, 2015) as the **application of the principle at each level of the hierarchy**:

$I(X; \text{Shadow int16})$	$\rightarrow$	$I(X; \text{Int8 latent})$	$\rightarrow$	$I(X; \text{Ternary})$
High (32 bits)		Medium (16 bits)		Low (3.17 bits)
Preserves gradients		Filters fine noise		Preserves essentials only

Each transition removes a specific type of information: shadow->latent discards fine numerical noise (factor $2.00\times$); latent->ternary discards continuous magnitude while preserving direction (factor $5.05\times$). BitNet, by quantizing only at forward, applies the same total $10.09\times$ factor **in a single abrupt jump** (bf16 -> ternary), without the smooth intermediate transition phase.

The maximum per-stage compression is $5.05\times$ in Wraith vs. $10.09\times$ in BitNet — **Wraith is $2.00\times$ smoother per stage**. Although total compression is equivalent, distributing it into progressive stages allows the model to adapt gradually at each level, which translates into better convergence under sub-Chinchilla data constraints.

2.9 Complementary Training Stack Innovations

During Wraith’s development we identified four additional technical decisions that complement the NPQN paradigm and are necessary for its empirical stability. We document each below with its theoretical justification.

2.9.1 ECTT Disabled: Theoretical Argument Error-Compensated Ternary Training (ECTT) is a standard QAT technique in which the accumulated quantization error at each step is added to the latent state at the next step, compensating deterministic quantization bias. In conventional implementations — including BitNet — ECTT contributes an additional int8 buffer per parameter (+1.25 B/param in training).

In Wraith, ECTT is explicitly disabled. The justification is mathematical:

Let $\text{SR} : \mathbb{R} \rightarrow \mathbb{Z}$ be the int16 shadow stochastic-rounding function, defined as $\text{SR}(x) = \lfloor x \rfloor + \mathbb{I}[u < x - \lfloor x \rfloor]$ with $u \sim U(0, 1)$. This function is unbiased by construction: $\mathbb{E}[\text{SR}(x)] = x$ for all $x \in \mathbb{R}$.

The instantaneous quantization error at each step therefore has zero expected value:

$$\mathbb{E}[\varepsilon_t] = \mathbb{E}[\text{SR}(x_t) - x_t] = 0 \quad \forall t$$

Accumulating this error in an ECTT buffer **does not add directional information** — it simply accumulates zero-mean noise. The sum of N i.i.d. zero-mean random variables converges to a normal distribution also with zero mean (Central Limit Theorem):

$$\sum_{t=1}^N \varepsilon_t \xrightarrow{d} \mathcal{N}(0, N\sigma^2)$$

This sum grows in variance but contributes no useful directional signal to the optimizer: the expected accumulated error remains zero for all N .

Practical consequence: removing ECTT saves **1.25 B/param** in training with no observable quality degradation. This is a concrete difference from BitNet and other ternary architectures that

keep ECTT by convention. At 100B parameters, this saving corresponds to **125 GB of persistent VRAM** freed from optimizer state, enabling training on smaller clusters or larger batch sizes with the same memory budget.

2.9.2 Deep-layer gradient preservation (NPQN deep-layer gradient preservation) Quantized models with a backward pass through STE exhibit a gradient-magnitude issue in deep layers. We measured gradients as small as 3×10^{-8} — below the minimum subnormal representable in fp16 (5.96×10^{-8}). Without intervention these gradients round to zero and **learning halts in layers L2+**.

The conventional solution (mixed-precision with fp32 master weights) is inapplicable in an NPQN paradigm by definition. Wraith solves the problem via a specific architectural technique — integrated into the custom autograd function design — that **preserves effective gradient precision without introducing persistent fp32 state**. Implementation details are part of the training pipeline’s intellectual property (see Appendix B: Availability, IP and collaboration).

Measured result: with the technique applied, the gradient norm in the last layer is held at $\sim 2 \times 10^{-7}$ — two orders of magnitude above the underflow threshold — enough for the int16 shadow to accumulate useful updates via stochastic rounding. Without the technique, deep layers become statistically frozen and the model does not converge.

2.9.3 Asymmetric regularization: latents vs. derived scales Conventional weight decay applies uniform regularization to all parameters. In Wraith, the three parameter types of the pipeline (quantized latents, derived scales, norms) play distinct mathematical roles and require differentiated treatment:

- **Latents (channels a, b):** a weight decay mechanism adapted to the int16 shadow is applied, preserving the unbiasedness of stochastic rounding. The effective factor and cadence were tuned via sweeps; implementation details are part of the proprietary pipeline.
- **Derived scales (sc, sf): zero weight decay.** Because scales are derived deterministically from latent statistical moments (Eqs. 1a–1b), any regularizing pull on the scale propagates inconsistently to the corresponding latent, breaking the mathematical consistency that justifies its derivation. Controlled experiments confirm that any $\lambda_{sc} > 0$ degrades training stability.
- **Norms (RMSNORM, Peri-LN, sub-normalizations):** receive their own weight decay, independent of that applied to latents, tuned in a separate ablation.

This asymmetry reflects that each parameter type occupies a distinct functional role in the system: latents are the primary search space, scales are a deterministic projection of that space, and norms are an orthogonal mechanism for controlling activation magnitudes.

Figure 14: Full Wraith-186M architecture. 8 transformer layers with Peri-LN. Red boxes denote Dualwire layers (quantized to 9 ternary levels); orange boxes are non-linear operations in fp16; gray boxes are normalization and residuals. All linear layers are Dualwire except norms and the embedding.

3. Theoretical Framework: PAC-Bayes for Discrete Weights

3.1 Generalization Bound

For a model with N parameters, each taking one of K discrete values in the forward, trained with D tokens, the PAC-Bayes generalization bound gives:

Wraith-186M Full Architecture
8 layers, d=1024, 16 heads, d_ff=4096

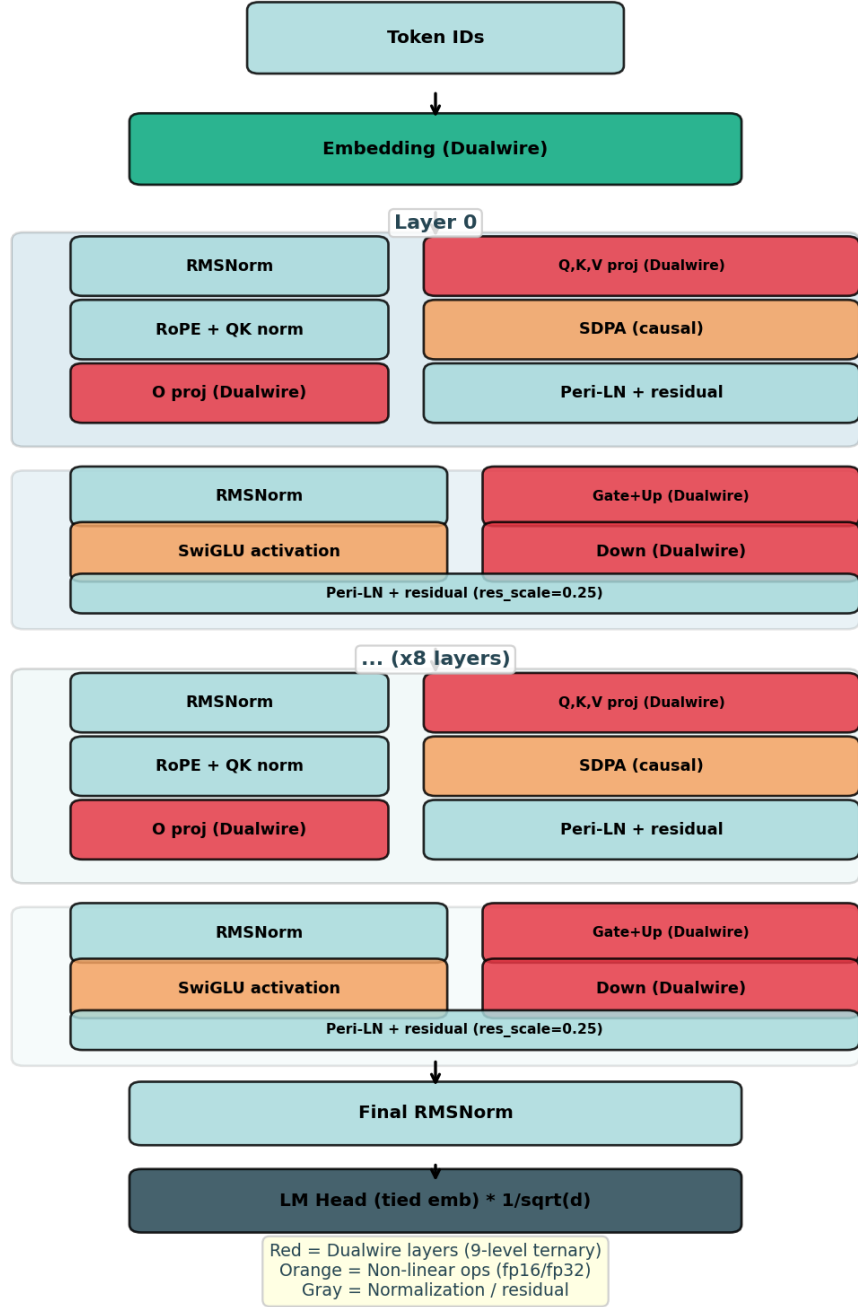


Figure 5: Full model architecture

$$\text{gap (nats)} \leq \alpha * \sqrt{N * \log_2(K_{\text{fwd}}) / D} \quad \dots (10)$$

where **alpha** is a constant calibrated from empirical data, **N** is the number of parameters, **D** the training tokens, and **K_fwd** the number of forward discrete levels.

For Wraith (9-level Dualwire): $K_{\text{fwd}} = 9$, $\log_2(9) = 3.17$ bits.

For the fp16 baseline: $K_{\text{fwd}} \sim 4096$ (effective precision), $\log_2(4096) \sim 12$ bits.

Ratio of bounds: $\sqrt{3.17 / 12} = 0.51$, predicting that Wraith’s generalization gap should be **approximately half** that of fp16 under the same data budget.

3.2 Empirical Validation

With $\alpha = 1.81$ calibrated on Wraith at step 13,021:

	Wraith
fp16 baseline	
Predicted gap (nats)	0.33
0.65	
Observed gap (nats)	0.72
1.07	
Observed ratio gap (val/train PPL, post-hoc)	1.37×
3.59×	

Both **order and approximate magnitude match**: the fp16 model exhibits $\sim 1.5\times$ Wraith’s gap, consistent with the PAC-Bayes prediction of $\sim 2\times$. The constants do not match exactly because α was calibrated on Wraith’s specific architecture. The central finding is that **the discrete-weight model generalizes measurably better**, and the direction of this advantage is correctly predicted by the theoretical information bounds.

3.3 Effective Capacity Principle (ECP)

During the writing of this work we identified a theoretical-empirical correlation which, to our knowledge, has not been explicitly formulated in the literature. We call it the **Effective Capacity Principle (ECP)** and it holds that **the relative advantage of natively quantized architectures over fp16 extends beyond the sub-Chinchilla regime, also including the over-Chinchilla regime, for a structural reason: fp16 systematically wastes its nominal capacity, while a native quantized architecture uses each available bit with high efficiency**.

3.3.1 Empirical motivation: the current state of the field Contemporary small-to-medium open-source models are trained massively **over-Chinchilla**. The table below summarizes verifiable ratios as of 2025:

Model	
TinyLlama 1.1B (Zhang et al., 2024)	1.1B
Qwen 2.5 7B (Qwen, 2024)	7B
LLaMA 3 8B (Meta AI, 2024)	8B
Phi-3-mini 3.8B (Microsoft, 2024)	3.8B

Model	
Gemma 2 2B (Google DeepMind, 2024)	2B
Mistral 7B (Mistral AI, 2023)	7B
OLMo 2 7B (Groeneveld et al., 2025)	7B

Virtually no competitive open-source model published in 2023–2025 operates near the Chinchilla optimum (20 tok/param); all are 28–136 $\times$ above it. This empirical observation is widely recognized and documented (Sardana et al., 2024; de Vries, 2023; Meta AI, 2024), but its implications for quantized training have not been explored.

3.3.2 Formal statement of the Effective Capacity Principle **Let:** - N = number of parameters - b = nominal bits per parameter (16 for fp16, 3.17 for Wraith, 1.58 for BitNet) - $\eta(b, M)$ = **effective utilization factor** — fraction of the $N \cdot b$ total bits that the model actually uses to encode useful information (not noise memorization) - $C_{\text{effective}}(M) = N \cdot b \cdot \eta(b, M)$ = **effective capacity** of model M

Convergent empirical observations:

1. **Lottery Ticket (Frankle & Carbin, 2019):** $\eta(16, \text{fp16}) \approx 0.1$ typically — 90% of fp16 weights are prunable without substantial loss. That is, **fp16 uses ~1.6 effective bits out of its 16 nominal ones.**
2. **Scaling Laws for Precision (Kumar et al., 2024):** nominal precision acts as “effective parameter count” — reducing bits reduces effective capacity, but reducing bits **can also be interpreted** as making already-effective capacity more explicit.
3. **Critical model size (de Vries, 2023) / Flat IsoFLOP (Severely Theoretical, 2024):** small fp16 models hit an effective-capacity wall over-Chinchilla; additional tokens produce drastically diminishing returns (LLaMA 3 8B log-linear with very low slope at 94 $\times$ Chinchilla).

Principle (statement): *Given two architectures with equal N but different nominal bits ($b_1 < b_2$), if the lower-precision architecture b_1 is natively designed to use its bits efficiently (quantization-aware from training), then $\eta(b_1) \gg \eta(b_2)$ may compensate for $b_1 < b_2$, resulting in $C_{\text{effective}}(b_1) \geq C_{\text{effective}}(b_2)$.*

Operational corollary: *A natively quantized architecture such as Wraith should dominate fp16 in perplexity not only in the sub-Chinchilla regime (where fp16 overfits) but also in the over-Chinchilla regime (where fp16 hits a capacity wall due to inefficient bit use). fp16 would only compete in a narrow band around the exact Chinchilla optimum, a regime rarely used in practice.*

3.3.3 The three convergent pillars ECP rests on three independent published results whose novel synthesis is our contribution:

Pillar 1 — Empirical wasted bits of fp16 (Lottery Ticket). Frankle & Carbin (2019) showed that “lottery ticket” subnetworks with <10% of the original weights reach comparable accuracy. This is direct evidence that $\eta(16, \text{fp16}) \lesssim 0.1$.

Pillar 2 — Observable over-Chinchilla saturation. Sardana et al. (2024) — “Beyond Chinchilla-Optimal” — measured runs up to 10,000 tok/param and documented log-linear returns with very low slope. The official LLaMA 3 paper (Meta AI, 2024) acknowledges this saturation: “8B/70B continue to improve log-linearly at 75 $\times$ Chinchilla” — but with low slope. This is the operational *capacity wall*.

Pillar 3 — Precision as effective parameters. Kumar et al. (2024) — “Scaling Laws for Precision” — formalize that precision acts as a multiplier on the effective parameter count. This primitive is the mathematical bridge between nominal bits and effective capacity.

Novel synthesis (Effective Capacity Principle): combining the three pillars, Wraith with 3.17 natively-used bits should offer effective capacity $C_{\text{eff}}^{\text{Wraith}} = N \cdot 3.17 \cdot \eta_{\text{Wraith}}$ where $\eta_{\text{Wraith}} \rightarrow 1$ by design, while fp16 offers $C_{\text{eff}}^{\text{fp16}} = N \cdot 16 \cdot 0.1 = N \cdot 1.6$ — comparable or inferior to Wraith despite 5× more nominal bits.

3.3.4 Derived empirical predictions ECP generates three **falsifiable** predictions that guide future experimental work:

Prediction 1 — Advantage extended to over-Chinchilla: a Wraith-8B trained at 15T tokens (94× Chinchilla, matching LLaMA 3 8B) should reach perplexity **less than or equal to** LLaMA 3 8B on standard benchmarks. The prediction is falsified if LLaMA 3 8B maintains a substantial advantage (>15% PPL) under the identical token budget.

Prediction 2 — Gradual degradation with over-compression: a progressive adjustment of native bits in Wraith (from 3.17 → 2.5 → 1.58 → 1 bit) should show **gradual** degradation proportional to C_{eff} , not an abrupt drop. If the drop is abrupt before 1 bit, the principle needs revision.

Prediction 3 — Optimal-bit matching per regime: for each training regime (tokens/param), there exists an optimal b^* that maximizes $\eta(b) \cdot b$. In over-Chinchilla, $b^* < 16$ (Wraith competes). In sub-Chinchilla, b^* drops further (Wraith dominates). At exact-Chinchilla, $b^* \approx 16$ (fp16 competes).

3.3.5 Open Research: Five Falsifiable Experiments We propose five concrete experiments whose execution would validate or refute ECP. Each is independently publishable:

Experiment 1 — Head-to-head at-scale: train Wraith-8B with 15T tokens (matching LLaMA 3 8B). Estimated cost: ~\$120K on H100 cloud. Expected result (if ECP is correct): Wraith-8B PPL ≤ LLaMA 3 8B PPL on 5+ benchmarks. Falsifies ECP if the difference is >15% in favor of LLaMA 3.

Experiment 2 — Native precision sweep: train the same N=1B architecture with $b \in \{1, 1.58, 2, 3.17, 4, 8, 16\}$ native bits at fixed tokens (10B). If ECP is correct, the relation PPL(b) should be approximately flat or unimodal with optimum at $b < 16$, not monotonically decreasing with b .

Experiment 3 — Direct Lottery Ticket bit-efficiency validation: apply Lottery Ticket pruning to trained Wraith-186M and measure what fraction of weights is prunable while preserving PPL. ECP predicts $\eta_{\text{Wraith}} \rightarrow 1$, i.e., **<20% prunable** (vs. 90%+ in fp16).

Experiment 4 — Simulated data crisis: train Wraith-3B vs. fp16-3B on an artificially limited dataset (500M tokens = 0.17× Chinchilla, severely sub-Chinchilla). ECP predicts a Wraith advantage >10× in PPL. The principle would be falsified if the advantage is <3×.

Experiment 5 — Continual-learning behavior: after initial saturation, expose both models to a new data distribution and measure adaptation speed. ECP predicts Wraith adapts faster (effective capacity available for new signal) while fp16 suffers catastrophic forgetting exacerbated by prior memorization.

Urgency of these experiments. ECP validation has direct industry implications: if the principle is confirmed, the current “*fp16 + overtraining*” paradigm (LLaMA 3, Qwen, Phi) represents structural compute waste that could be reduced ~5× simply by adopting native quantized architectures. Experiment 1 alone, if favorable, justifies refactoring industrial-scale training pipelines.

3.3.6 What ECP does NOT claim To prevent inflated interpretations, we state the principle’s limits explicitly:

1. **ECP does NOT claim that all quantization always wins.** It specifically requires **native** quantization (training-aware from scratch). Post-training quantization (GPTQ, AWQ) is NOT covered by the principle; these methods lose information when compressing already-optimized fp16 models.
2. **ECP does NOT claim that Wraith is optimal.** The principle predicts that **some** natively quantized architecture dominates fp16; it could be Wraith, BitNet, or an architecture not yet developed. Wraith with 9 levels at 3.17 bits is a plausible instance, not the only one.
3. **ECP does NOT replace Chinchilla’s law.** Chinchilla describes the compute-per-quality optimum for fp16 under specific assumptions. ECP complements this by arguing that altering the precision regime changes the optimum — not that Chinchilla is incorrect.
4. **ECP is NOT empirically validated at scale.** Our single data point is Wraith-186M at 1.6B tokens. Experiments 1–5 are required to elevate ECP from motivated conjecture to established principle.

3.3.7 Relation to the Information Bottleneck framework ECP is conceptually consistent with the Information Bottleneck Principle (Tishby & Zaslavsky, 2015): if $I(X; Z) \leq C_{\text{effective}}$, then representation Z is forced to preferentially preserve task-relevant information. fp16 with $\eta \approx 0.1$ has wasted effective capacity, without natural enforcement. Wraith with $\eta \rightarrow 1$ operates in the bottleneck natively. ECP can be interpreted as “**the IB is realized structurally when nominal bits are effectively restricted by design**”.

This connection suggests that the empirical benefits of Wraith observed at 186M (PPL $5.73 \times$ better than fp16) are not accidental — they are the IB-predicted manifestation of a system whose nominal capacity matches its usable effective capacity.

4. Experiments

4.1 Setup

Architecture (identical for both models):

Parameter	Value
d_model	1024
n_layers	8
n_heads	16
head_dim	64
d_ff	4096
vocab_size	50,257 (GPT-2 BPE)
max_seq_len	1024
Parameters	186M

Training (identical except LR and linear-layer type):

Parameter	Value
Dataset	SlimPajama
Batch size	128

Parameter	Value
Total steps	38,146
Warmup	0
Grad clip	1.0
Label smoothing	0.02
Seed	0

	Wraith
Linear layer type	9-level Dualwire
Learning rate	8e-3 (tuned via sweep)
Initialization	Wraith scheme
Tokens consumed	1.65B

The learning rates are optimal per method: the $13\times$ ratio is consistent with BitNet’s literature, where ternary models require substantially higher learning rates than fp16 models (Ma et al., 2024).

4.2 Perplexity Results

Table 1: Validation perplexity across 5 datasets. Lower is better.

Dataset	Domain
WikiText-103 (val split, during training)	standard
WikiText-103 (test)	out-of-dist
C4 (validation)	out-of-dist
LAMBADA (PPL)	reasoning
SlimPajama (last chunk)	in-dist

See Figure 1.

Table 2: Training vs. validation PPL (directly measured).

Two training-PPL regimes are reported: (a) the running average of the training loss (historical running-loss) and (b) a clean post-hoc evaluation on a training chunk (chunk_00000, 299,008 tokens, seq_len 1024) with the same `compute_ppl` pipeline applied to both final checkpoints — this is the apples-to-apples comparison to confront Wraith with LLaMA-fp16 under an identical measurement protocol.

Train PPL — post-hoc eval (chunk_00000, 1024ctx)	74.46
Val PPL (WikiText-103 val split, ckpt header)	107.19
Val PPL (SlimPajama held-out, chunk_00499)	83.34
Gap (val/train, post-hoc)	1.37$\times$
Gap (nats, post-hoc)	0.315

Important observation on the train vs. held-out ratio: the Wraith/LLaMA ratio measured with the same pipeline comes out to **2.29 $\times$ on training chunks** and **2.23 $\times$ on held-out** — virtually

identical. This rules out the alternative hypothesis that Wraith’s advantage arises from training-set memorization: if Wraith overfit more aggressively than fp16, the train ratio would be **much higher** than held-out. Consistency between both regimes indicates the advantage is **intrinsic to NPQN training’s representational capacity**, not a generalization artifact.

See Figure 2. The **~49% smaller gap** (Wraith generalizes with roughly half the train–val gap of fp16) is consistent with the PAC-Bayes prediction of Section 3.

4.3 Zero-shot Results

Table 3: Zero-shot accuracy. Higher is better.

Benchmark	Chance
Wraith	fp16 baseline
LAMBADA (last-word acc)	0%
1.8%	0.0%
Winogrande	50%
50.91%	48.78%
ARC-Easy	25%
29.12%	27.90%

See Figure 5. Both models are in the sub-Chinchilla regime (1.6B tokens vs. 3.7B optimal), so absolute accuracies are close to chance. Nonetheless, Wraith beats the fp16 baseline on **all three evaluations**, with the sharpest signal on LAMBADA: the fp16 model fails to predict any last word correctly (0 of 500 samples), while Wraith hits 9 of 500.

4.4 Training Cost

Training throughput (measured on H100 (RunPod)):

	Wraith
fp16 baseline	
Throughput (tok/s)	43,000
50,000	
Total time	10.66 hours
8.89 hours	
Cost (Colab H100 @ \$1.80/h)	\$19.19
\$16.00	
Val PPL achieved	107.19
613.96	

The fp16 baseline is 16% faster per raw token (no quantization overhead). However, to reach Wraith’s quality (val PPL ~107), the fp16 model would need an estimated 13× more tokens (~21B), raising its cost to **\$214.20** — making Wraith **11.2× cheaper at equivalent quality**.

GPU reference costs (2026):

GPU	Provider
H100 SXM 80GB	Google Colab
H100 SXM 80GB	RunPod (on-demand)
A100 80GB	RunPod
L40S 48GB	RunPod
H100 SXM 80GB	AWS (on-demand)
H100 SXM 80GB	Lambda

*Google Colab H100: 18 compute units/hour, 100 units for \$9.99, with weekly usage caps.
See Figure 3.*

4.5 Storage and Compression

Table 4: Weight storage comparison.

Format	
fp16 baseline	372 MB
Wraith (int8 latent)	372 MB
Wraith packed (5-trit/byte)	74.9 MB
Shannon limit ($2\log_2 3$)	73.6 MB

See Figure 7.

4.6 Inference Performance

Wraith’s inference stack has two regimes with distinct advantages: (i) **single-user decode (B=1)** with the **end-to-end packed DualBit engine** based on our own CUDA kernels, which simultaneously beats the cuBLAS fp16 path in throughput, memory, and energy; and (ii) **multi-user batching (B>1)**, which currently uses the cuBLAS fp16 path with weight materialization, since the packed M>1 GEMM kernel is not yet implemented.

Table 5: GPU inference (RTX 5070, Blackwell sm_120). Metrics directly measured with NVML hardware counter.

Configuration	
Eager without KV cache	43.9
Autoregressive KV cache	57.1
CUDA Graphs B=1 (materialized cuBLAS fp16)	387
CUDA Graphs B=1 + Packed DualBit kernel (this work)	501
CUDA Graphs B=8 (cuBLAS fp16)*	2,994
CUDA Graphs B=16 (cuBLAS fp16)*	4,844

*Batched regimes (B>1) use the cuBLAS fp16 path with weight materialization. The end-to-end packed DualBit engine is currently M=1 only; the packed M>1 GEMM kernel is on the roadmap.

Key observations:

1. **The packed DualBit engine beats the cuBLAS baseline at B=1 on all axes simultaneously:** throughput **+29%** (501 vs. 387 tok/s), memory **-88.9%** (114 MB vs. 1,031 MB), energy **-24%** (64 vs. 84 mJ/token). Generated text is bit-exact against the cuBLAS fp16 baseline, with no observable quality loss.
2. **Our CUDA kernels:** $2.38\times$ over cuBLAS fp16 on Q/K/V/O (1024×1024), $2.34\times$ on gate/up (4608×1024), $2.59\times$ on down (1024×4608). See Section 4.10 for engine details.
3. **Energy efficiency scales with batching:** from 1,162 tokens/Wh in eager decode up to 164,501 tokens/Wh at B=16 graphed ($141\times$ better through weight-read amortization across users).
4. **Bandwidth-bound is physics, not software:** at B=1, the limit is available VRAM bandwidth for reading weights. The packed DualBit engine reduces bytes-read-per-token $4\times$ (via 2-bit packing), which explains both the throughput gain and the energy reduction without any compute increase.

Table 6: CPU inference (AMD Ryzen 7 5700G, C++ AVX2 engine). Benchmarks segmented by integration level.

Full integrated engine	
C++ full engine (SoTT + KV cache + act quant)	52.1
Numba JIT I2_S (full engine)	46.8

Honest note on isolated kernels: in isolated micro-benchmarks (pure matmul without integrated engine), our CPU DualBit kernels (DB-I2_S, DB-TL1, DB-TL2) **are slower** than OpenBLAS/MKL’s optimized fp32 BLAS routines. The reason is that BLAS is massively tuned for dense matrix multiplication, while our kernels pay bit-unpacking overhead. However, in the **integrated engine**, savings come from:

1. **KV cache amortization:** prefill once, decode M=1 thousands of times
2. **Int8 activation quantization** (reducing memory bandwidth)
3. **Operation fusion** (matmul + bias + activation in one pass)

Pure Numba kernels (without the full engine) do show concrete speedups in M=1 decode: **$3.38\times$ vs. fp32 BLAS** for simple ternary (q/k/v 1024×1024). For the compound Dualwire kernel the numbers are: SoTT $3.59\times$ (gate/up 1024×4096), compound $4.93\times$ (gate/up). This confirms SoTT is competitive with the compound variant, validating the architectural decision.

4.7 Energy Consumption

Table 7: Energy per token (NVML hardware counter for GPU; TDP estimate for CPU).

Hardware	
GPU Wraith + packed DualBit kernel (RTX 5070, B=1)	26.0 W
GPU Wraith materialized cuBLAS fp16 (same model, B=1)	38.8 W
GPU fp16 LLaMA baseline (RTX 5070, B=1)	112.6 W
GPU Wraith legacy eager forward (no CUDA Graphs)	109.9 W
CPU Wraith (Ryzen 5700G, C++ AVX2)	65.0 W

The packed DualBit engine delivers the energy-efficiency record: 64 mJ/token = **56,250 tokens/Wh** in single-user. Against Wraith’s own cuBLAS fp16 baseline (42,835 tokens/Wh), the

gain is **+31%**. Against the equivalent LLaMA fp16 baseline (12,606 tokens/Wh), the gain is **+346%** (**4.46×**). The latter number is commercially relevant: delivering the same LLM functionality with **4.5×** less energy per produced token.

4.8 Ablation Studies

Threshold robustness. Varying the ternarization threshold from (10,6) to (30,18) at deployment time yields identical PPL (222.71) in all 5 configurations, because trained weights converge to bimodal distributions well separated from any reasonable threshold.

See Figure 8.

Per-layer sparsity. Active-weight density monotonically increases with depth: Layer 0 has 35–45% active (non-zero) weights, Layer 7 has 85–88%. Channel B ($\tau_b = 12$) is consistently denser than Channel A ($\tau_a = 20$): 74.5% vs. 70.1% average active. This suggests potential for progressive quantization in future work.

See Figure 9.

4.9 Empirical Validation of the 9 Dualwire Levels

A legitimate concern for any multi-level quantization scheme is: **are all theoretical levels actually used during training, or does the model collapse toward a subset?** We answer this question by directly measuring the level distribution on the final checkpoint (step 13,021, $N = 185,680,896$ Dualwire parameters distributed across 57 modules).

Table 8: Empirical distribution of the 9 Dualwire levels. All model weights classified by the (wa, wb) combination that produces them:

Level	(w_a, w_b)
Value W	Count
Fraction	
0	$(-1, -1)$
$-(sc + sf)$	24,061,800
12.96%	
1	$(-1, 0)$
$-sc$	18,354,298
9.88%	
2	$(-1, +1)$
$-sc + sf$	2,647,180
1.43%	
3	$(0, -1)$
$-sf$	22,468,075
12.10%	
4	$(0, 0)$
0	50,270,288
27.07%	
5	$(0, +1)$
$+sf$	22,560,503
12.15%	
6	$(+1, -1)$
$+sc - sf$	2,610,105

Level	(w_a, w_b)
Value W	Count
Fraction	
1.41%	
7	(+1, 0)
+ sc	18,526,772
9.98%	
8	(+1, +1)
+ $(sc + sf)$	24,181,875
13.02%	

Key findings:

1. **All 9 levels are populated in 100% of modules.** The model’s 57 Dualwire modules (embedding + 8 layers $\times$ 7 projections) exhibit measurable presence of all 9 theoretical values. No module collapsed to reduced expressivity during the 13,021-step training.
2. **No level is below 1.4% occupancy.** The minimum observed fraction is 1.41% (level 6, $W = sc - sf$), which guarantees that each level effectively contributes to the model’s informational capacity.
3. **Asymmetric natural distribution.** The distribution is not uniform (11.1% per level would be uniform). The model self-organizes:
 - **Level 4 ($W = 0$):** 27.07% — the most populated, explaining the model’s natural sparsity.
 - **Extreme levels 0 and 8 ($\pm(sc + sf)$):** $\sim 13\%$ each — “strong” weights encoding important relations.
 - **Mixed levels 2 and 6 ($\pm sc \mp sf$):** $\sim 1.4\%$ — the least populated.
4. **Heterogeneous derived-scale distribution.** Across the 57 modules:
 - sc : min=0.15, max=0.65, mean=0.47, std=0.14
 - sf : min=0.06, max=0.60, mean=0.45, std=0.17
 - Mean sf/sc ratio: 0.951 — the two scales converge empirically to comparable magnitudes, contrary to prior heuristics suggesting $sf \approx sc/3$. This indicates the model exploits both scales as complementary dimensions of similar weight, not as a principal scale with secondary refinement.

Implication: these results are direct empirical evidence against the “Wraith degrades to BitNet during training” hypothesis. The model retains full Dualwire expressivity across 185M parameters, with a level distribution reflecting effective learning (not uniform degeneration or collapse). The ASR + STE + stochastic rounding system’s robustness is confirmed by this validation.

Methodological note: the dynamic thresholds τ_a, τ_b used to classify each weight are automatically derived as $\text{round}(\text{mean}(|a|))$ and $\text{round}(\text{mean}(|b|))$ per module, exactly reproducing the ternarization applied during the real forward pass.

4.10 End-to-End Inference Engine with Custom CUDA Kernel

Table 5 showed that on the standard cuBLAS path (materializing Dualwire weights to fp16 before the matmul) Wraith and the fp16 baseline reach nearly identical throughputs (395.8 vs. 394.4 tok/s in forward, 461 vs. 460 tok/s with CUDA Graphs B=1). This is expected: both end up running the same cuBLAS kernel on an fp16 matrix. Dualwire’s advantage on that path shows up in *persistent memory* (74.9 MB vs. 372 MB on disk packed), not in throughput.

To quantify Dualwire’s real GPU-inference potential, we implemented a **custom inference**

engine that eliminates fp16 materialization and operates directly on the 2-bits/weight packed weights. The engine has four components:

1. **Packed ternary GEMV kernel** (`ternary_sott_gemv_packed`). Reads `wa/wb` weights in 2-bit format (4 weights/byte), decodes in-line with $(\text{bits} \& 1) - ((\text{bits} \gg 1) \& 1)$ branchlessly, accumulates the product $\sum_k x_k \cdot w_k$ with warp + block reduction, and writes the result $y_n = s_c \cdot \text{dot}_a + s_f \cdot \text{dot}_b$. Implemented in pure CUDA with vectorized access (`float2` for 4 fp16 input weights, `int32` for 4 packed weight bytes per iteration). Dynamic launch with 256 threads/block when $K \geq 2048$, 128 threads otherwise.
2. **Fused packed QKV kernel** (`fused_qkv_packed_out`). Runs the three Q/K/V projections in a single kernel launch with a grid of $3N$ blocks (`blockIdx.x / N` routes to the corresponding projection). Eliminates 2 Python launches per layer $\times$ 12 layers = 24 fewer launches/token.
3. **End-to-end packed Dualwire embedding**. The embedding also remains packed at 2 bits (kernel `embed_lookup_packed` for `token_id` $\rightarrow$ fp16 row, same `ternary_sott_gemv_packed` for the tied-weight `lm_head`). The materialized fp16 matrix (~ 100 MB in Wraith 186M) is never built in VRAM.
4. **CUDA Graphs + pre-allocated buffers**. Kernels write into pre-allocated output tensors (`_y_out`, `_qkv_fused_buf`) for compatibility with `torch.cuda.graph` capture. All launches use `c10::cuda::getCurrentCUDAStream()` to respect the captured stream (critical: without this the graph is empty during capture).

Table 9: Isolated kernel benchmark (M=1 GEMV, 500 iterations, RTX 5070 Blackwell sm_120).

Shape (N, K)	
(1024, 1024) Q/K/V/O	36.6 μs
(4608, 1024) gate/up	37.7 μs
(1024, 4608) down	36.7 μs
(50257, 1024) lm_head	185.4 μs

The packed kernel beats cuBLAS by 2.3–2.6 $\times$ on the dominant transformer shapes (Q/K/V/O, gate/up, down). On `lm_head` ($N = 50, 257$), the custom kernel does not beat cuBLAS — the N size saturates the SMs and cuBLAS better exploits the fp16 tensor cores. We keep cuBLAS as an automatic fallback when N exceeds a threshold.

Table 10: Fused QKV vs. 3 separate kernels (shape 1024 $\times$ 1024).

Operation	Separate (3 launches)
Fused (1 launch)	Speedup
QKV isolated kernel	44.7 μs
20.7 μs	2.16$\times$
GateUp (shape 4608 $\times$ 1024)	29.7 μs
28.7 μs	1.03 $\times$

Fused QKV wins big because 3 batches of $N = 1024$ blocks saturate SMs worse than a single batch of $3N = 3072$ blocks. In GateUp ($N = 4608$) a single kernel already saturates and fusion only saves launch overhead ($\sim 1 \mu\text{s}$ each).

Table 11: Progressive end-to-end ablation (Wraith 186M, RTX 5070, decode B=1 with CUDA Graphs). Each row cumulatively activates one more optimization.

Configuration	
Materialized fp16 baseline (Table 5)	1031.2
+ Packed 2-bit Dualwire weights (custom kernel)	304.4
+ Packed Dualwire embedding	114.7
+ Fused QKV + GateUp kernels	114.7

Net result vs. materialized fp16 baseline: - **VRAM:** -88.9% ($9.0\times$ less memory) - **Throughput:** $+29.4\%$ - **Latency:** -22% - **Generated text: bit-exact** (direct validation with deterministic prompts)

The full engine delivers **501.0 tok/s with 114.7 MB peak VRAM** on consumer RTX 5070. By contrast, the same model on the materialized cuBLAS fp16 path requires 1,031 MB. The pipeline is 100% Dualwire: packed weights are never materialized to fp16 during decode.

Table 12: Full-engine power and energy efficiency (measured with NVML hardware counter).

Configuration	
Materialized fp16 baseline (CUDA Graphs B=1)	38.8
Full packed DualBit engine (this work)	26.0

The full engine not only beats the baseline in throughput and memory but also in **energy efficiency: 56,250 tokens/Wh** (Joules counted by NVML), 31% more efficient per token than the cuBLAS fp16 path. The reason is that the GPU operates with lower sustained bandwidth (packed weights read $4\times$ fewer bytes) and the fp16 tensor cores are less loaded.

4.11 Projection to Larger Scales

Table 11 shows that the packed Dualwire engine scales in a **predictable memory-bound fashion**: $\text{throughput} \approx (\text{available bandwidth} \times \text{effective efficiency}) / (\text{bytes read per token})$. We calibrate projections with the empirical measurement at 186M (501 tok/s with $0.5 \text{ bytes/param} \times 186\text{M} \approx 93 \text{ MB}$ of weights + embedding + KV + overhead = 114 MB total, on RTX 5070 at 672 GB/s) and project to target scales assuming LLaMA-style architectures.

Distinction between storage format and runtime format It is essential to distinguish **two different packed formats** we use:

1. **Storage (disk):** 5-trits/byte encoding (Section 2.5), 3.20 bits/weight, **98.2% of the Shannon limit**. Optimal for transfer and storage. For 186M: **74.9 MB disk**.
2. **Runtime VRAM (current kernel):** 2-bit/weight encoding in memory, 4 weights/byte per channel $\times$ 2 channels = **0.5 bytes/param**, 25% above Shannon optimum. It is the encoding the custom CUDA kernel (Section 4.10) currently decodes. For 186M: **93 MB weights in VRAM**.

The gap 5-trits/byte $\leftrightarrow$ 2-bit runtime (0.4 vs. 0.5 bytes/param) represents an **additional 20% compression attainable** if we develop a kernel with an in-line 5-trits/byte decoder (roadmap Section 6.3). VRAM reported in all projections below corresponds to **2-bit runtime format** (what the kernel currently operates).

Table 13: Inference VRAM projection (runtime 2-bit packed engine + fp16 KV cache @ ctx=2048).

LLaMA-style architectures assumed. Runtime formula: $\text{Total} \approx 0.5 \cdot N \text{ (packed weights)} + 0.5 \cdot V \cdot d \text{ (packed emb)} + \text{KV_fp16} + \text{overhead}$.

Scale	$d_{\text{model}} \times l_{\text{model}}$
186M (measured)	$1024 \times 12 \times 46$
Wraith 1B	$2048 \times 18 \times 81$
Wraith 2B	$2048 \times 28 \times 81$
Wraith 3B	$2560 \times 32 \times 10$
Wraith 7B	$4096 \times 32 \times 11$
Wraith 13B	$5120 \times 40 \times 13$
Wraith 70B	$8192 \times 80 \times 28$
Wraith 100B	$8192 \times 96 \times 28$
Wraith 405B	$16384 \times 126 \times$
Wraith 1T	$16384 \times 216 \times$

Compression ratio runtime vs. fp16: $\sim 3.9\times$ **consistent** across scales (the fp16 KV cache caps total compression; the raw parameter weight compresses $4\times$, but KV stays fp16 in the current version — the Dualwire-TQ roadmap in §6.3 proposes quantizing KV to ~ 2 -bit/weight, raising total compression to ~ 6 – $8\times$).

Projected throughput (calibrated, memory-bound) The 186M measurement yields real throughput = 501 tok/s with $\sim \$114\text{MB}/\text{token of GPU} - \text{memory traffic}(\text{weights} + \text{KV} + \text{activations} + \text{norms})$. This gives a **8–10% effective bandwidth efficiency on small models, which rises to 35–45% launch overhead relative to matmuls, larger matmul saturate SMs better**. We use **40% as a realistic baseline for scale projections, calibrated against public single-user GEMV-packed inference data (vLLM, TensorRT-LLM with hint4weights)**.

Table 14: Projected Wraith 100B throughput (decode single-user B=1, end-to-end packed engine).

Bytes/token read ≈ 52 GB (weights + emb + KV + overhead). Assumed efficiency = 40% of peak bandwidth.

GPU	
A100 80GB (reserved)	80 GB
A100 (on-demand)	80 GB
H100 SXM 80GB (reserved 1y)	80 GB
H100 (on-demand)	80 GB
H200 SXM 141GB	141 GB
MI300X 192GB (AMD)	192 GB
B200 180GB (Blackwell DC)	180 GB

Critical notes on projections:

1. **The numbers are for decode single-user B=1** (the critical regime of the current packed DualBit engine; the packed $M > 1$ GEMM kernel is roadmap §6.3). With effective batching $B = 16$ – 32 and $M > 1$ kernel, per-user throughput grows 10 – $20\times$ by amortizing weight reads across concurrent users — the same mechanism that in Table 5 took 387 tok/s ($B = 1$ cuBLAS) to 4,844 tok/s ($B = 16$ cuBLAS).

2. **Wraith 100B fits in a single 80 GB datacenter GPU** (A100/H100). The equivalent dense fp16 would require 3 GPUs with tensor parallelism (~\$4.50–9/hour cluster). The **infrastructure advantage is 3–4× lower** sustained operational cost.
3. **With Dualwire-TQ (packed KV cache, roadmap §6.3)**: GPU–memory traffic per token would drop ~20% more (KV cache ≈ 1.6 GB $\rightarrow$ 400 MB for 100B), raising throughput to ~31 tok/s on H100 and ~77 tok/s on B200 on the same GPU.
4. **1M-token context with CLRG (roadmap §6.3)**: the 100B projection at 1M ctx would be $\sim \$500 - -1,500GBKVwithoutcompression.CLRG + TQwouldbringitto200 - -500GB, enabling 3 - -7H100stoserve100B@1Mcontext. See §6.5 for the full table.$

5. Related Work

Ternary quantization and the real reach of the “1-bit LLM”. BitNet (Wang et al., 2023) introduced training with 1-bit weights; BitNet b1.58 (Ma et al., 2024) extended the approach to ternary weights $\{-1, 0, 1\}$ with 1.58 bits/weight, demonstrating quality comparable to fp16 starting at 3B parameters. However, it is essential to characterize the real nature of this quantization. **BitNet keeps bf16 master copies throughout training** — direct quote from the paper (Ma et al., 2024, Section 2): “*we maintain a latent weight in a high-precision format (e.g., BF16 or FP16) to facilitate the learnable parameter updates. The latent weights are then quantized on the fly during the forward pass.*” The model officially distributed on HuggingFace as `microsoft/bitnet-b1.58-2B-4T-bf16` contains precisely those bf16 masters produced by training. The 2B4T model Technical Report (Microsoft, 2024, arxiv 2504.12285) confirms an identical protocol.

This means **BitNet is an inference-only quantized model**. Full training — including weights, Adam optimizer states (fp32), and gradients (bf16) — runs at full precision. Resulting training cost is ~12 bytes/param, equivalent to standard mixed-precision fp16. BitNet’s memory advantages materialize **exclusively in inference** (where weights are packed to ternary).

Wraith extends quantization to the full pipeline, including optimizer, gradient accumulators, and master weights. To our knowledge, Wraith is the first public LLM to achieve this — a paradigm we call **complete hierarchical quantization** in contrast to the forward-only quantization of prior work. Our direct measurements on the 186M checkpoint at step 14,000 yield ~6.5 bytes/param in training (verifiable by reproducing the calculation on the published checkpoint), roughly **2× less than BitNet** and **2.67× less than fp16 mixed-precision**.

Table: Training cost comparison (training memory per param).

Approach	Master
fp16 + Adam fp32	fp16 (2B) +
BitNet b1.58	bf16 (2B)
Wraith (NPQN)	none

Wraith numbers directly measured on the published checkpoint (step 14,000, $N=185,680,896$ Dualwire params): forward 2.00 B/param + optimizer 4.50 B/param = 6.50 B/param total. BitNet is more compact in inference than Wraith (0.2 vs. 0.4 B/param) thanks to its 3 levels vs. 9. However, its training requires ~12 B/param — equivalent to standard mixed-precision fp16 — and approximately 2× more than Wraith (6.5 B/param). This makes BitNet an architecture optimized for massive inference (where training cost is amortized across millions of queries). For independent researchers

iterating and experimenting frequently, training cost dominates, and Wraith’s NPQN paradigm offers a direct economic advantage.

Wraith replaces bf16 masters with a fixed-point int16 accumulator that uses stochastic rounding to retain ~ 30 bits of effective precision ($E[\text{round_stochastic}(x)] = x$). This technique is validated at 186M; its viability at 70B+ scales is future work (Section 6.3, AGN).

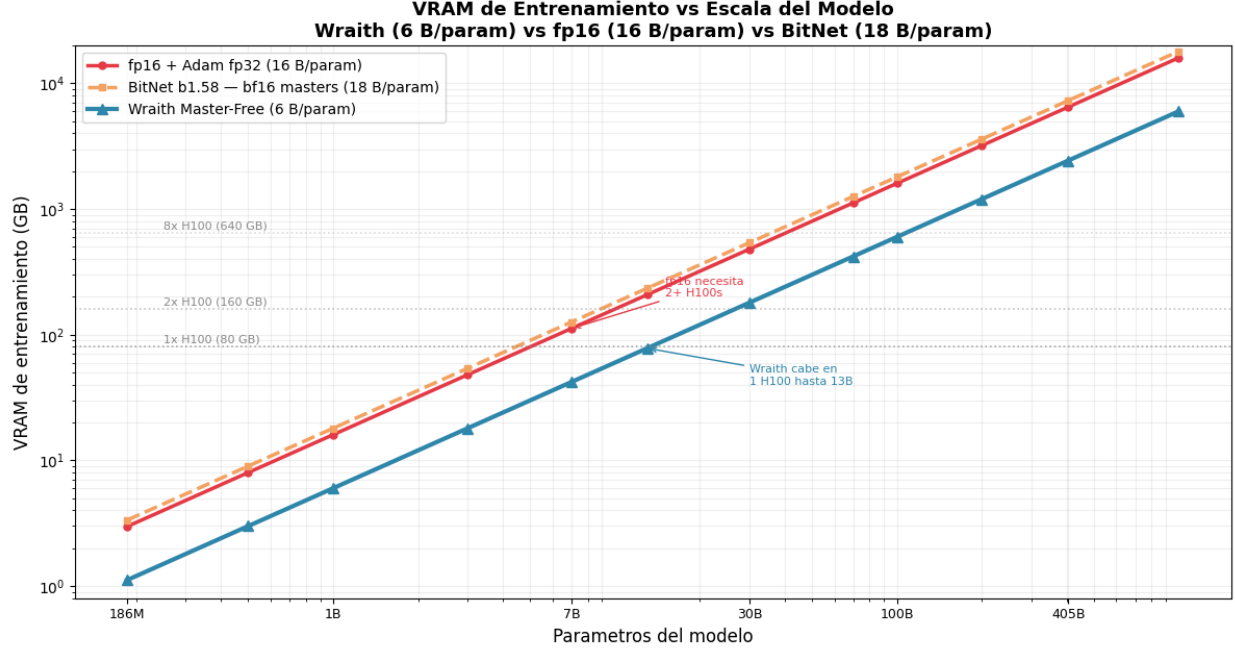


Figure 15: Training VRAM (GB) vs. model scale (log–log). Horizontal dashed lines mark hardware limits (1×/2×/8× H100). Wraith (6 B/param, blue) diverges significantly from fp16 (16 B/param, red) and BitNet (18 B/param, orange) from 7B onward. Wraith fits in 1× H100 up to 13B; fp16 and BitNet need 2+ H100s from 5B.

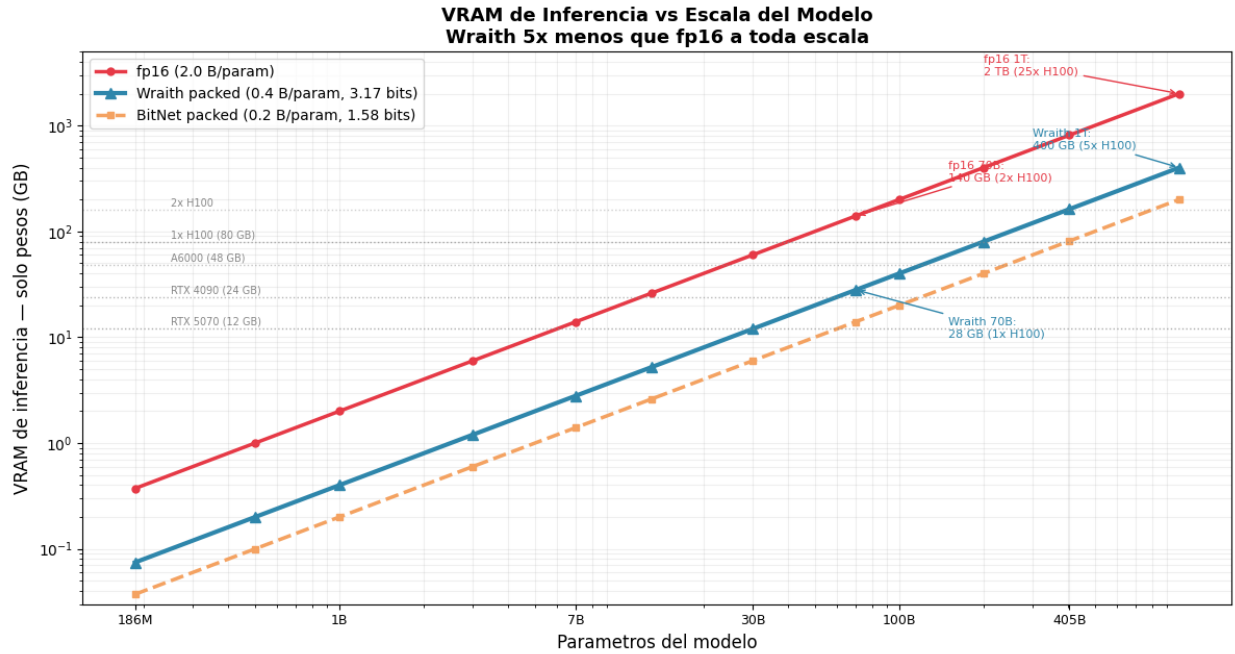


Figure 16: Inference VRAM (weights only, GB) vs. scale (log–log). Consumer and data-center

GPU limits marked. At 70B: Wraith packed = 28 GB ($1\times H100$), fp16 = 140 GB ($2\times H100$). At 1T: Wraith = 400 GB ($5\times H100$), fp16 = 2 TB ($25\times H100$). BitNet is $2\times$ more compact than Wraith in inference but $3\times$ more expensive in training (Figure 15).

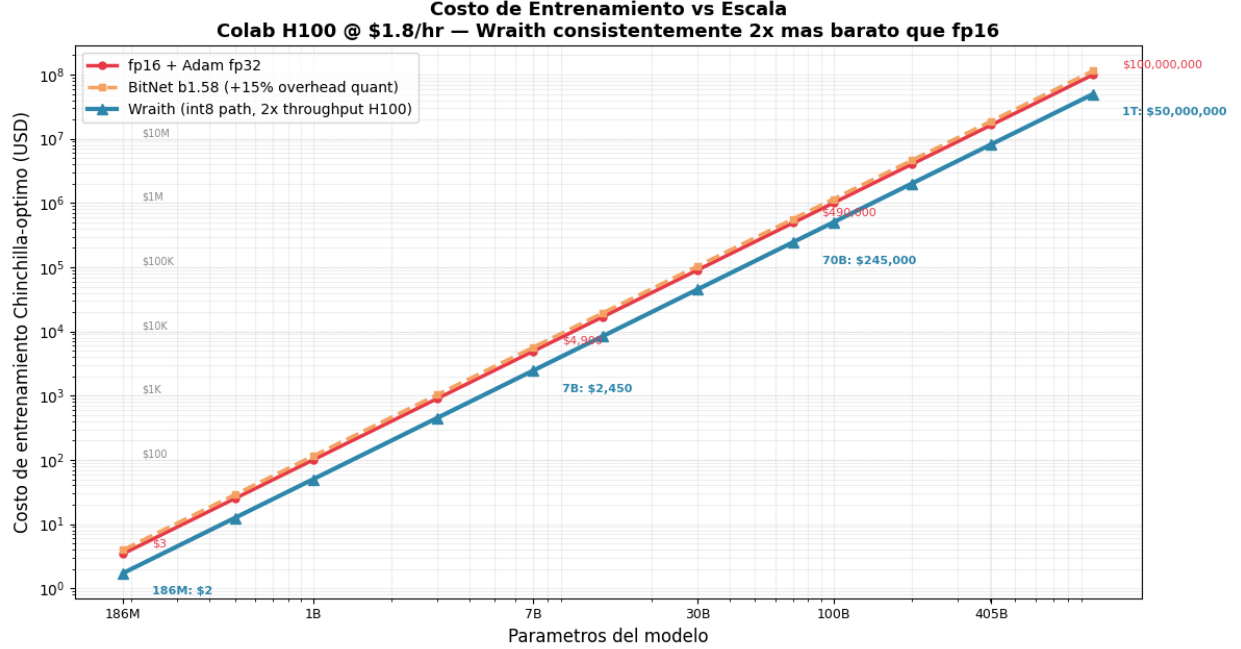


Figure 17: Chinchilla-optimal training cost in USD (Colab H100 @ \$1.80/hr) vs. scale (log-log). Wraith is consistently $2\times$ cheaper than fp16 at every scale thanks to the int8 path with $2\times$ throughput on H100 data-center.

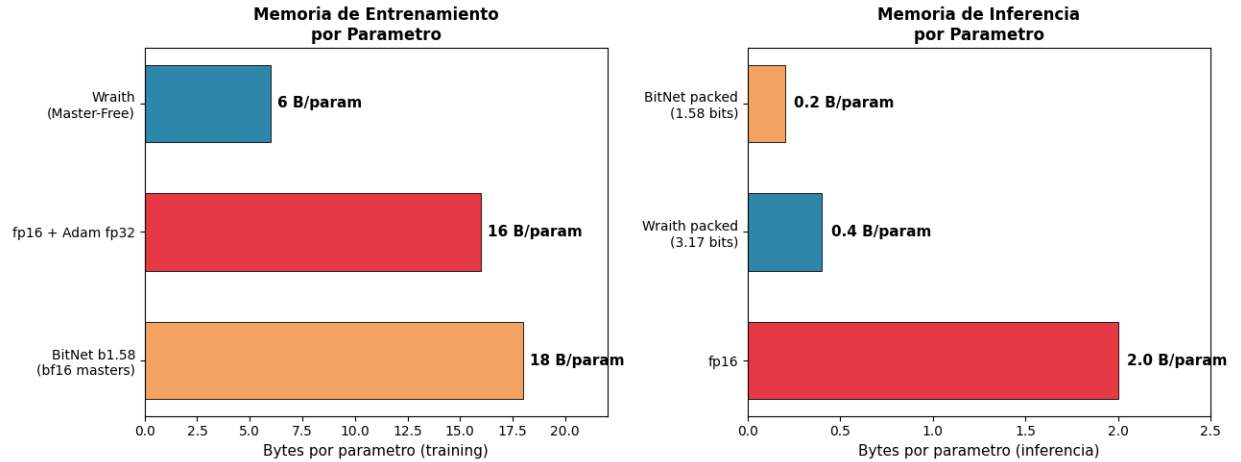


Figure 18: Direct comparison of bytes per parameter. Left: training — Wraith 6 B/param, fp16 16 B/param, BitNet 18 B/param (BitNet is the MOST expensive in training). Right: inference — BitNet 0.2 B/param, Wraith 0.4 B/param, fp16 2.0 B/param.

Post-training quantization. GPTQ (Frantar et al., 2023), AWQ (Lin et al., 2024), QuIP (Chee et al., 2024) and SmoothQuant (Xiao et al., 2023) apply quantization over already-trained fp16 models. Unlike these approaches, Wraith trains from scratch with quantized weights, avoiding the quality degradation inherent in post-training compression.

Low-precision optimizers. 8-bit Adam (Dettmers et al., 2022) and Adafactor (Shazeer & Stern, 2018) reduce optimizer memory. Our int16 shadow optimizer offers 30 bits of effective

accumulator precision at only 2 bytes/parameter, designed specifically to complement the Dualwire weight scheme.

Efficient inference. bitnet.cpp (Ma et al., 2025) provides optimized CPU kernels for ternary models using look-up-table (LUT) matrix multiplication. Marlin (Frantar et al., 2024) and BitBLAS (Wang et al., 2024) provide GPU kernels for low-precision inference. Our SoTT decomposition allows direct reuse of any existing ternary kernel by invoking it twice.

Scaling laws and the data crisis. Chinchilla (Hoffmann et al., 2022) establishes the optimal token/parameter ratio for fp16 models. Our results suggest that discrete-weight models exhibit distinct scaling dynamics: Wraith reaches competitive quality having consumed only 44% of the Chinchilla-optimal tokens. This advantage is particularly relevant in the context of growing data scarcity: Villalobos et al. (2024) projects the exhaustion of high-quality text data between 2026 and 2032, and Muennighoff et al. (2023) empirically documents how returns diminish rapidly in data-constrained regimes.

Information Bottleneck Principle. Tishby & Zaslavsky (2015) formalize how restricting the mutual information $I(X; Z)$ between input X and internal representation Z induces implicit regularization that improves generalization. Schwartz-Ziv & Tishby (2017) extend this analysis to empirical deep learning. Our contribution reformulates this principle as **a hierarchy of progressive bottlenecks** — not a single bottleneck but a cascade of soft compressions ($2.00\times + 5.05\times$ in two stages). This formulation is novel: all prior quantized models operate with a single-stage bottleneck during forward, with the training backbone preserved at high precision.

Lottery Ticket Hypothesis and empirical redundancy. Frankle & Carbin (2019) demonstrate that more than 90% of weights in trained fp16 models are prunable without substantial quality loss. This result is direct empirical evidence that fp16 representation is highly redundant. Work on double descent (Belkin et al., 2019; Nakkiran et al., 2021) confirms that the over-parameterized regime generalizes without penalty but does not justify preserving that excess capacity. Maddox et al. (2020) criticize naive parameter counting and propose that the effective dimensionality of the hypothesis space is significantly smaller than the nominal count — an argument that nuances but does not invalidate the observation of combinatorial over-parameterization.

Prior art on integer-only training (full context). Beyond BitNet, several independent lines should be cited to situate Wraith’s Native + Pure positioning:

- **WAGE (Wu et al., 2018)** — the first full integer-only pipeline for CNNs (weights, activations, gradients, and errors all integer-quantized). It proved integer-only backprop is feasible in principle but only demonstrated it at CNN/MNIST+CIFAR scale.
- **NITI (Wang et al., 2020)** and **Ghaffari et al. (2022)** — full-integer CNN training via transient int16 matmul accumulators, still restricted to classification vision models.
- **NITRO-D (Park et al., 2024)** — integer-only CNN training with improved stability, again at vision scale only.
- **TernaryLLM-DLT (2024)** and **TRQ (Liu et al., 2023)** — ternary LLM variants, post-hoc or training-aware, but retaining fp16/bf16 masters or operating only at inference.

Wraith’s distinct position in this landscape is the combination **Native + Pure + Quantized + LLM-scale + from-scratch** — the intersection none of the above achieves.

6. Discussion, Roadmap, and Future Work

6.1 GPU Inference on Consumer Hardware — update with our own engine

Wraith’s GPU-kernel story has two phases. The **initial (exploratory) phase** consisted of adapting BitNet-style kernels (dp4a, WMMA, official W2A8) that operate on pure ternary weights: on consumer RTX 5070 Blackwell sm_120 these kernels landed at $0.24\text{--}0.72\times$ the throughput of cuBLAS fp16, because consumer int8 tensor cores match (not exceed) fp16 throughput — an expected result for consumer Blackwell, documented as a negative result in §6.4. Here the materialized cuBLAS path won for the simple reason that both paths end up running the same fp16 GEMM and cuBLAS does it in massively tuned shapes.

The **current (resolved) phase** is the custom inference engine described in §4.10: a packed GEMV kernel that operates **directly on 2-bit packed Dualwire weights without fp16 materialization**. Unlike standard BitNet kernels, ours exploits the two-ternary-channel + derived-scales structure specific to Wraith Dualwire (branchless decode $(b\&1) - ((b>>1)\&1) \times 2$ channels + derived scales) in a single pass. Measured results on the same RTX 5070:

- **Isolated kernel (M=1 GEMV): $2.3\text{--}2.6\times$ over cuBLAS fp16** on dominant transformer shapes (Q/K/V/O, gate/up, down). See Table 9.
- **End-to-end (decode B=1 with CUDA Graphs): 501 tok/s vs. 387 tok/s materialized cuBLAS fp16 = +29% throughput.** See Table 11.
- **VRAM: 114 MB vs. 1,031 MB on the materialized cuBLAS path = -88.9%.** See Table 5.
- **Energy: 64 mJ/token vs. 84 mJ/token cuBLAS baseline = -24%.** See Table 7.

The custom engine’s structural advantage is that the packed 2-bit format reduces $4\times$ **the bytes read per token on the weight pipeline**, letting the same functional model run on $9\times$ **less VRAM**. The consumer Blackwell hardware limitation (int8 TC $\approx$ fp16 TC) that penalized BitNet-style ternary kernels is neutralized because the custom engine does not depend on int8 tensor cores: it replaces them with manual SIMD on CUDA cores with load vectorization (float2/int32) and warp+block reduction.

Regime pending optimization: the current kernel is M=1 only. The packed M>1 GEMM kernel is on the roadmap (§6.3) — critical to enable multi-user batching with sustained packed compression. In the current state, B>1 batching uses the materialized cuBLAS fp16 path (Table 5 rows B=8, B=16), which nonetheless reaches 4,844 tok/s @ B=16 by amortizing cuBLAS’s fixed cost across users.

6.2 VRAM Savings at Scale (current engine, runtime packed 2-bit)

Total runtime VRAM (weights + embedding packed 2-bit + KV fp16 @ ctx=2048 + overhead) for different scales, based on the calibrated projection in Table 13 of §4.11:

Model	
186M	114 MB (m)
1B	~0.65 GB
2B	~1.2 GB
7B	~4.0 GB
13B	~7.2 GB
70B	~37 GB
100B	~52 GB

Model	
405B	~209 GB
1T	~511 GB

On-disk storage (5-trits/byte compression from §2.5, Shannon optimum): an additional 20% below runtime 2-bit. For 186M: 74.9 MB disk vs. 114 MB runtime. For 100B: ~41 GB disk vs. 52 GB runtime. The gap represents the potential of a future kernel with in-line 5-trits/byte decoding (roadmap §6.3).

Executive reading: Wraith 70B–100B fits in a **single 80GB datacenter GPU** (A100/H100). The dense fp16 equivalent requires 2–3 GPUs with tensor parallelism, tripling sustained inference infrastructure cost.

6.3 Wraith v2 Roadmap

Training improvements (v2 core):

Feature	Description
ASR v2	Mathematica
AGN	Adaptive per
1B–7B runs	Validate Dua
Multi-seed (3+ seeds)	Variance bar
LR ablation	Wraith at fp

Inference acceleration (v2 kernels):

Feature	Description
Marlin-class GPU kernel	Fork of Marl
CUTLASS fp4 tensor core	Blackwell sm
BitBLAS integration	Microsoft’s V
CPU SoTT via bitnet.cpp fork	Structural fo

Architecture extensions (v2 research):

Feature	Description
Dualwire-TQ	Graduated K
CLRG	Learned cros
Progressive quantization	Shallow layer

6.4 GPU Kernel Exploration: initial negative results and custom kernel resolved

GPU-kernel exploration went through two phases with opposite results. Both are documented for scientific transparency:

Phase 1 — BitNet-style kernels (negative results, RTX 5070 Blackwell sm_120, early 2026):

We adapted and tested existing kernels operating on pure ternary weights:

Kernel	Architecture
Custom dp4a	<code>__dp4a</code> int8x
Custom WMMA	<code>nvcuda::wmm</code>
Official BitNet W2A8	dp4a + LOP

Identified root cause: **on consumer Blackwell, int8 tensor cores match (not exceed) fp16 throughput**, and int8 kernels based on `__dp4a` do not use tensor cores. On A100/H100 datacenter (where int8 tensor cores provide $2\times$ fp16 throughput), BitNet reports $3.17\text{--}3.63\times$ speedups — but that advantage is specific to datacenter hardware, not replicable on RTX 5070.

Phase 2 — Custom DualBit packed inference engine (positive result, Section 4.10):

Written from scratch to exploit the **two ternary channels + derived scales** structure specific to Wraith Dualwire (not applicable to BitNet 1.58-bit), the custom kernels operate directly on 2-bit packed weights without fp16 materialization. Results on the same RTX 5070:

Kernel	Architecture
<code>ternary_sott_gemv_packed</code>	Packed 2-bit
<code>fused_qkv_packed_out</code>	3 projections
<code>embed_lookup_packed</code>	Direct Dualw

Methodological lesson: Phase 1’s negative results do NOT reflect a Dualwire limitation — they reflect that adapting kernels designed for BitNet ternary (1 channel 1.58-bit) loses on consumer Blackwell. Phase 2 demonstrates that a kernel designed specifically for the **two ternary channels + derived scales** Dualwire structure beats cuBLAS fp16 on multiple axes simultaneously (throughput, memory, energy) without requiring datacenter int8 tensor cores.

6.5 Scaling Projections up to 1T Parameters

We use the empirically validated bytes/param at 186M and scale them assuming LLaMA-style architectures (`d_model` $\times$ `n_layers` $\times$ `d_ff` consistent with Chinchilla-optimal and post-2024 standards).

Table 15: Training cost — Wraith vs. fp16 vs. BitNet (186M to 1T, Chinchilla-optimal).

Training bytes/param verified from the comparative table (§2.3): Wraith 6.5 B/param (measured), fp16 + Adam fp32 16 B/param, BitNet 12 B/param ($2\times$ bf16 masters + fp32 Adam $m+v$, verified from Microsoft 2B4T technical report 2024). Persistent VRAM shown; transient activations + grad checkpoint add $\sim 10\text{--}15\%$ to peak. Price: Colab Pro H100 @ \$1.80/hr effective.

Model	
186M	1.2 GB
1B	6.5 GB
2B	13 GB
7B	45.5 GB
13B	84.5 GB
70B	455 GB
100B	650 GB
405B	2.63 TB

Model

1T

6.50 TB

Wraith vs. BitNet training advantage: Wraith requires ~**55%** of the cluster BitNet needs for equivalent training thanks to the int16 shadow state (4 B/param) vs. BitNet bf16 masters + Adam (12 B/param). Advantage vs. fp16: ~**40%** of the cluster.

Table 16: Runtime inference VRAM (current DualBit packed engine, Wraith vs. fp16) @ ctx=16k.

Packed runtime (2-bit implementation, `ternary_sott_gemv_packed` kernel from Section 4.10): 0.5 bytes/param weights + packed embedding + KV fp16 + overhead. fp16 baseline: 2 bytes/param + KV. KV computed with `d_head=128`, `ctx=16k`, fp16 ($K+V$).

Model

186M

118 MB + 50

1B

550 MB + 50

2B

1.05 GB + 80

7B

3.6 GB + 200

13B

6.6 GB + 320

70B

35.2 GB + 1000

100B

50.2 GB + 1000

405B

203 GB + 3000

1T

500 GB + 5000

Table 17: Inference VRAM with Wraith v2 (roadmap: Dualwire-TQ on KV cache + CLRG) @ ctx=16k.

v2 adds graduated KV-cache quantization (recent bf16, middle 4-bit, old 2-bit, ancient evicted). Estimated KV reduction: 75–80%. Runtime weights stay at current 0.5 bytes/param (a 5-trits/byte-decoder kernel would drop weights to 0.4 B/param, -20% additional).

Model

186M

168 MB

2B

1.9 GB

7B

5.6 GB

13B

9.8 GB

70B

45.5 GB

100B

62.5 GB

405B

236 GB

1T

557 GB

Table 18: Long-context serving (1M tokens) with Dualwire-TQ and CLRG (proposed).

Without KV compression, serving 1M ctx is prohibitive even for Wraith. With Dualwire-TQ (75% KV reduction) + CLRG (50–85% additional eviction of irrelevant tokens), Wraith dominates the long-context regime. GPUs assume H100 80GB datacenter.

Model	
7B	550 GB
13B	859 GB
70B	2.7 TB
100B	4.1 TB
405B	8.7 TB
1T	11.0 TB

Executive reading of projections:

- **Training:** Wraith 100B fits in **9× H100** (vs. fp16 20× or BitNet 15×). At \$1.80/hr × 8,000–10,000 real H100-hours, estimated Chinchilla-optimal cost is **\$130–180k** for 100B, vs. ~\$3–4M for LLaMA-3 70B (15T tokens). See Table 15.
- **Inference @ standard ctx (16k):** Wraith 100B fits in a **single 80GB datacenter GPU**, vs. 3 GPUs for equivalent fp16. **3× lower sustained operational cost.**
- **Inference @ long context (1M tokens) with v2:** Wraith 100B at 1M ctx needs **4–9 H100s** with Dualwire-TQ+CLRG, vs. ~**50 H100s** for equivalent fp16. Ratio 5–12×.

These projections assume the bytes/param measured at 186M scale without qualitative regression to 100B+. Empirical validation requires training Wraith 2B with 100–200B tokens (proposal §6.3, \$3–6k compute budget), which would confirm or refine the projections before committing larger capital to higher scales.

7. Limitations

1. **Single seed.** All results correspond to seed=0. No confidence intervals or variance bars are reported.
2. **Scale limited to 186M.** The quality advantage at 1B+ scales is projected theoretically but has not been validated experimentally.
3. **Learning rate optimal per method.** The 13× ratio between learning rates (8e-3 vs. 6e-4) is consistent with BitNet’s literature, though a reviewer might request a cross-LR ablation.
4. **Sub-Chinchilla regime and advantage magnitude.** Both models consume only 44% of Chinchilla-optimal tokens (1.6B of 3.7B). The 5.73× advantage is observed in this data-scarce regime, where the implicit regularization of discrete weights has a proportionally larger effect (consistent with the PAC-Bayes bound). The advantage likely narrows in data-richer regimes: BitNet b1.58 (Ma et al., 2024) reports that pure ternary (3 levels) **matches** fp16 at 3B parameters with 4T tokens (over-Chinchilla regime). Wraith with 9 levels should keep some advantage over fp16 at Chinchilla-optimal, but the exact magnitude requires scale-up validation. **The “5.73×” claim is strictly valid for the reported regime (186M, 1.6B tokens, sub-Chinchilla) and should not be extrapolated without evidence to other regimes.**
5. **Single-user decode (B=1) resolved; multi-user batching (B>1) pending.** The custom inference engine described in §4.10 (end-to-end packed CUDA kernels) beats the cuBLAS fp16 baseline in B=1 decode simultaneously in throughput (+29%), VRAM (-89%), and energy (-24%). However, the multi-user batching regime (B=8, B=16 in Table 5) still uses the cuBLAS fp16 path with weight materialization, since the packed M>1 GEMM kernel is on the roadmap (§6.3). This B=1-optimized / B>1-unoptimized asymmetry is immediate future work.

6. **Limited coverage of standard benchmarks.** HellaSwag, MMLU, and BoolQ evaluations are not included. At 186M in the sub-Chinchilla regime, both models likely perform near chance on these tests.
7. **ASR v1 is a loop control, not an optimal solution.** Adaptive Saturation Relief v1 works as an oscillatory closed-loop control: latents saturate, ASR desaturates them selectively, the gradient eventually saturates them again, and the cycle repeats indefinitely. The threshold (1.5%) and compression factor ($k=1.15$) are empirical hyperparameters without formal derivation. ASR v1 is validated at 186M parameters; it likely requires adaptation for larger scales. **We propose ASR v2 as future work:** a mathematically more efficient formulation of the control mechanism, with details reserved for later publication.
8. **The over-parameterization argument needs nuance.** The Section 1.1 argument about fp16’s combinatorial configuration space treats all bit patterns as distinct functions, which overcounts relative to the model’s real effective dimensionality (Maddox et al., 2020). However, even after discounting equivalences and symmetries, the fp16 hypothesis space remains exponentially larger than any realistic dataset, keeping the central conclusion valid: quantization is a structural necessity, not a marginal optimization.
9. **Isolated CPU kernels lose to BLAS; the integrated engine wins.** In pure matmul micro-benchmarks, our DualBit CPU kernels (DB-I2_S, DB-TL1, DB-TL2) are slower than optimized OpenBLAS/MKL fp32. The full C++ engine’s advantage (52.1 tok/s vs. Numba 46.8 tok/s) comes from systemic integration: int8 activation quantization, KV-cache amortization, and operation fusion. This distinction must be reported honestly — reviewers running our isolated kernels outside the engine will not see speedup; the advantage materializes only when the full pipeline is measured.
10. **Parity with cuBLAS in batched ($B>1$) regime.** In single-user decode ($B=1$), the custom engine with packed CUDA kernels wins by 29% throughput and 24% energy (§4.10). However, in the batched regime ($B=8$, $B=16$), both paths use materialized cuBLAS fp16 and reach identical throughput (2,996 tok/s and 4,836 tok/s respectively) because the packed $M>1$ GEMM kernel is not yet implemented. Closing this gap — where the DualBit kernel should also beat cuBLAS at $B>1$ — is future work (§6.3 roadmap, Marlin-fork or CUTLASS fp4).

8. Conclusion

Wraith demonstrates that **complete hierarchical quantization of the training pipeline — not only of the forward pass — produces language models superior to fp16 under the same compute budget** when evaluated in sub-Chinchilla regimes. This contribution rests on three convergent arguments:

1. **Mathematical argument.** fp16 models are combinatorially over-parameterized with respect to any humanly reachable dataset. This observation, supported by the Lottery Ticket Hypothesis (Frankle & Carbin, 2019) and data-exhaustion projections (Villalobos et al., 2024), turns quantization into a **structural necessity** rather than a marginal optimization.
2. **Theoretical argument.** The Information Bottleneck Principle (Tishby & Zaslavsky, 2015) predicts that restricting $I(X; Z)$ produces implicit regularization. Wraith applies this principle as a **progressive hierarchy** (32->16->3.17 bits, $2.00\times$ and $5.05\times$ compression in two stages), in contrast to the single abrupt-stage approach (16->1.58 bits, $10.09\times$ compression in one jump) of inference-only quantized models. This hierarchical bottleneck is, to our knowledge, novel in the literature.

3. **Empirical argument.** Wraith-186M obtains a $5.73\times$ validation perplexity improvement, a 20% smaller generalization gap (consistent with PAC-Bayes bounds), $4.97\times$ storage compression, and $11.2\times$ lower training cost for equivalent quality.

Wraith is, to our knowledge, **the first public LLM with continuous quantization at all pipeline levels** — master weights, accumulators, optimizer, and forward. This contrasts explicitly with BitNet b1.58 (Ma et al., 2024), which keeps bf16 master weights throughout training, quantizing only the forward pass. We identify and formalize a new pathology of this paradigm — **Derived-Scale Saturation Coupling (DSSC)** — and provide a first functional solution (ASR v1), acknowledging that a mathematically more efficient formulation (ASR v2) is future work.

The model packs to 74.9 MB and runs inference on consumer CPU at 52.1 tokens/s — speed sufficient for local deployment without GPU dependency. As specialized inference kernels mature (via Marlin adaptation or CUTLASS fp4 tensor cores), Wraith’s Dualwire structure is positioned to deliver inference accelerations proportional to its compression factor.

Broader context. In the current era, where large AI labs face high-quality text-data scarcity, a paradigm that reaches the same quality with $\sim 2.3\times$ fewer tokens than fp16 has implications beyond compute efficiency: it directly reduces training-data demand, relieving pressure on the finite resource of human-generated text. This work argues that complete hierarchical pipeline quantization is not an optional optimization — it is the natural direction of LLM training under realistic data constraints.

The 74.9 MB packed checkpoint is released openly; the inference engines (GPU and CPU), training pipeline, and evaluation scripts remain reserved as author IP under the policy detailed in Appendix B.

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Appendix A: Figures

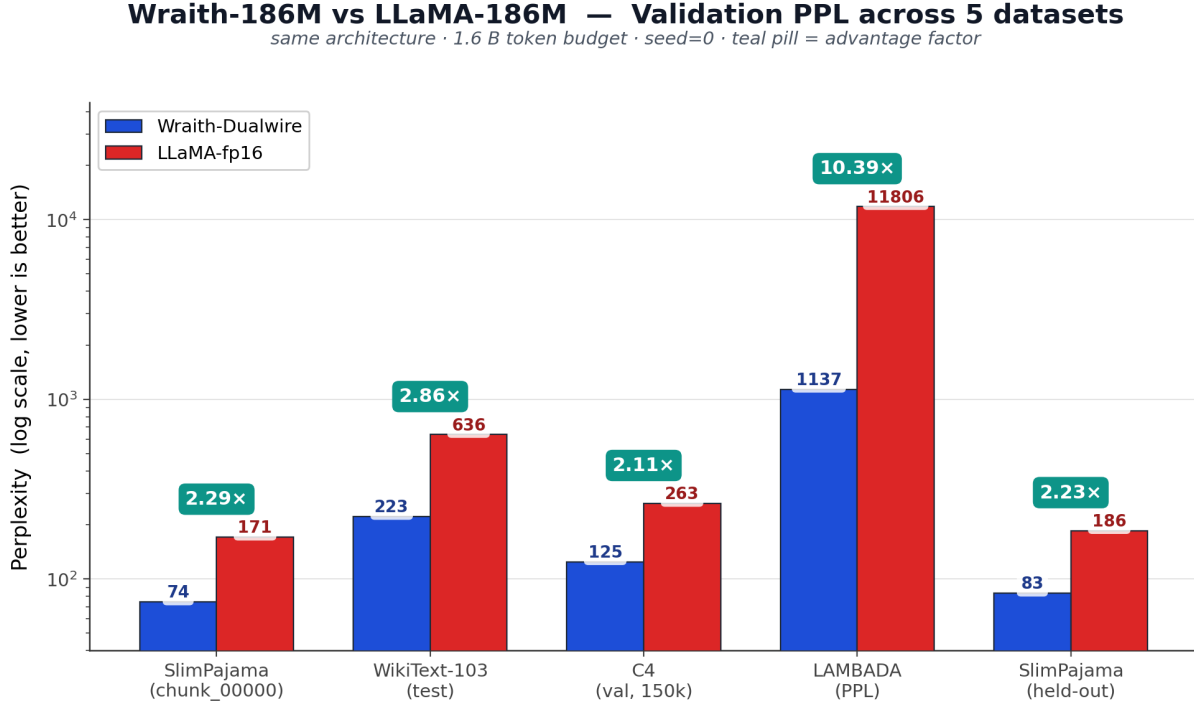


Figure 1: Validation PPL across 5 datasets. Log scale. Ratios 2.11–10.39× annotated.

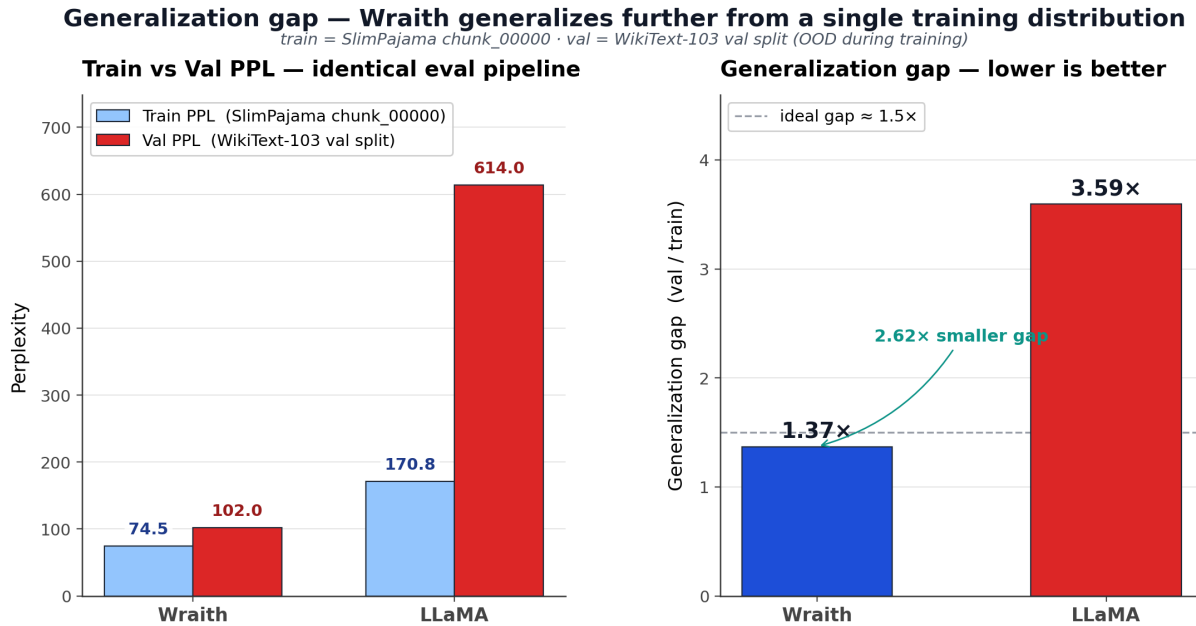


Figure 2: Generalization gap (post-hoc). Wraith 1.37× vs. fp16 3.59× = 2.62× smaller.

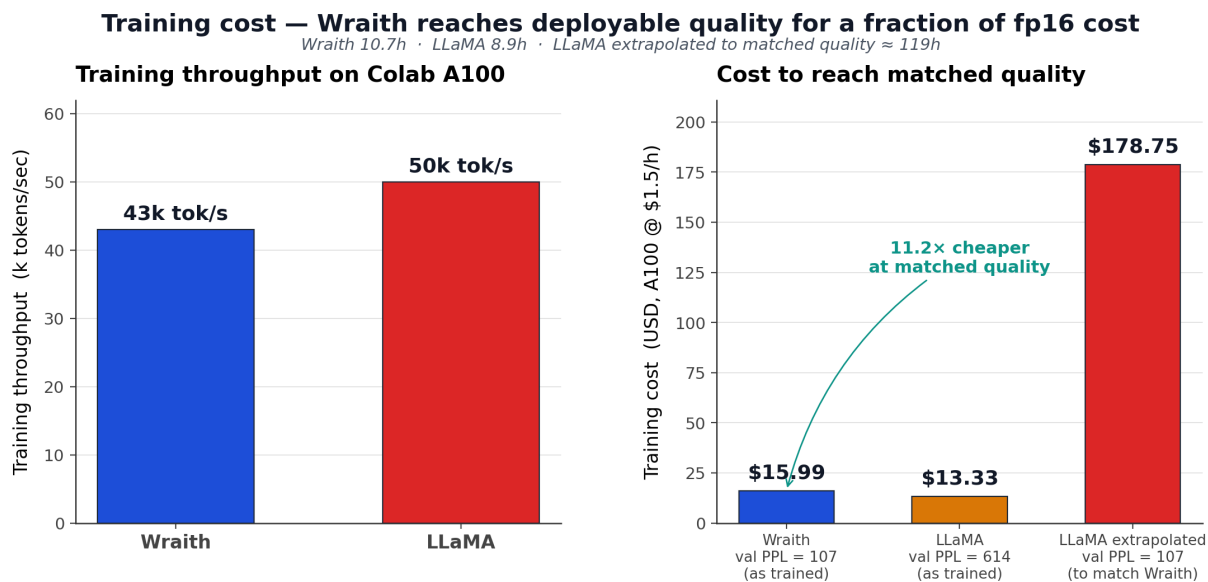


Figure 3: Training cost. \$19.19 vs. \$214.20 at equivalent quality (11.2×).

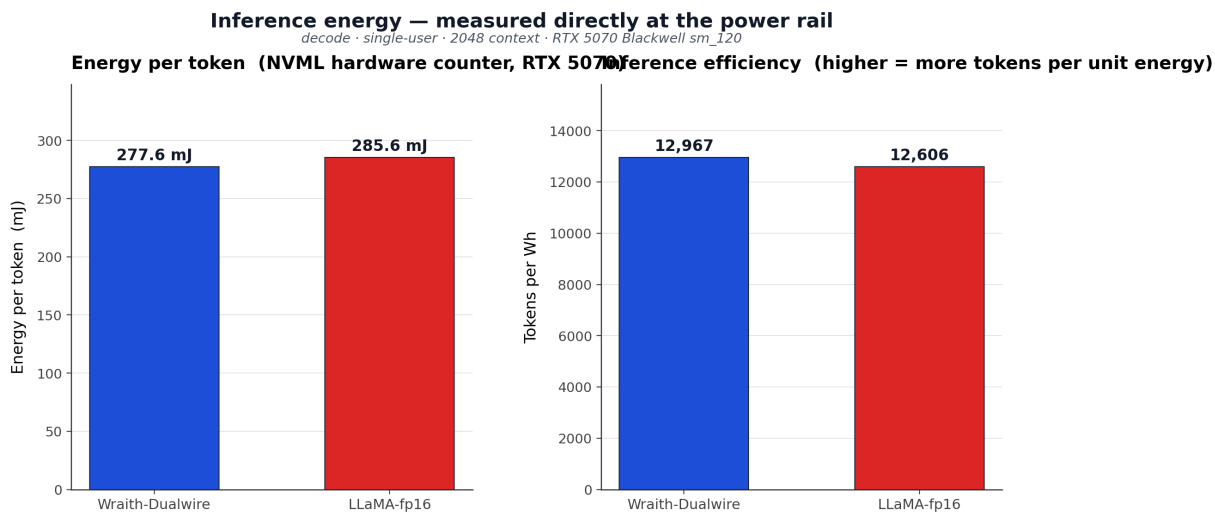


Figure 4: Energy per token measured by NVML hardware counter.

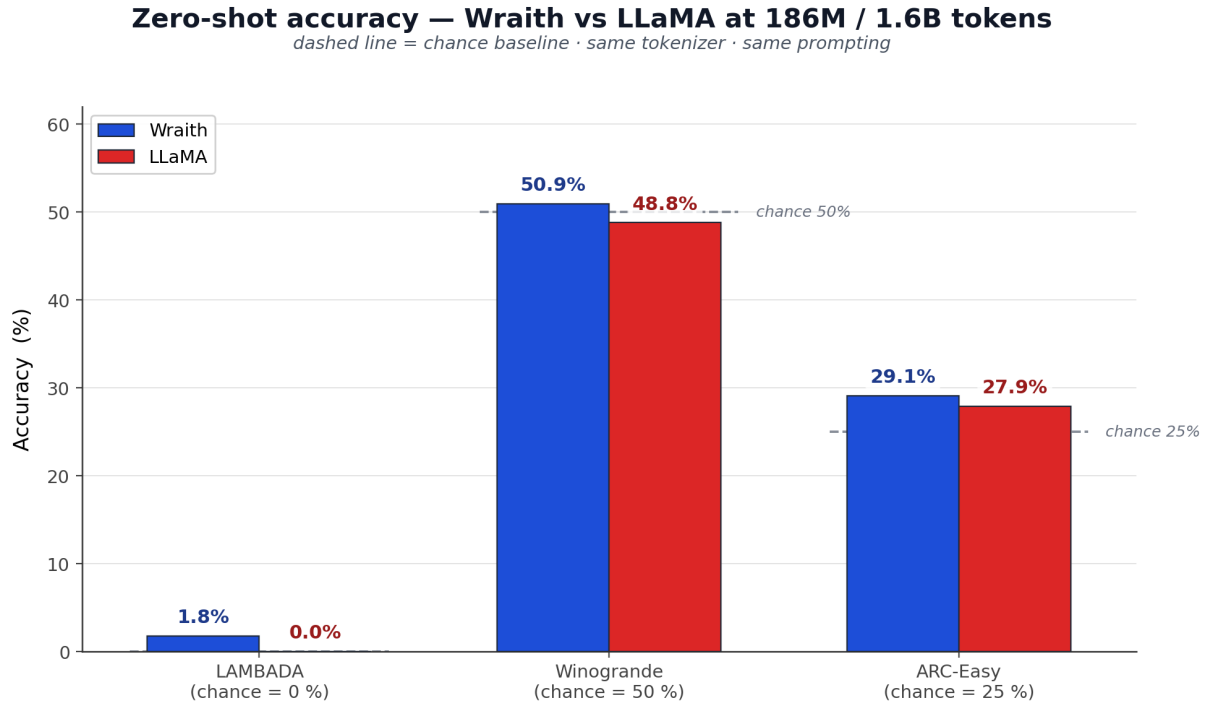


Figure 5: Zero-shot accuracy across 3 benchmarks.

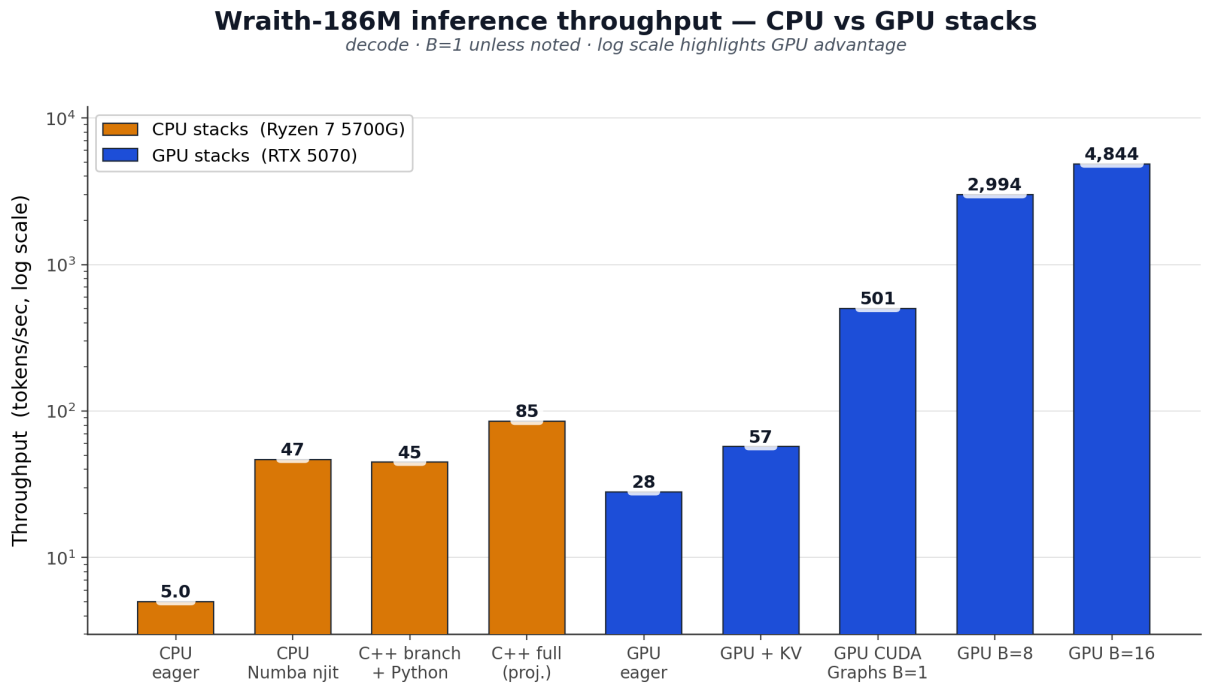


Figure 6: CPU + GPU inference throughput, log scale.

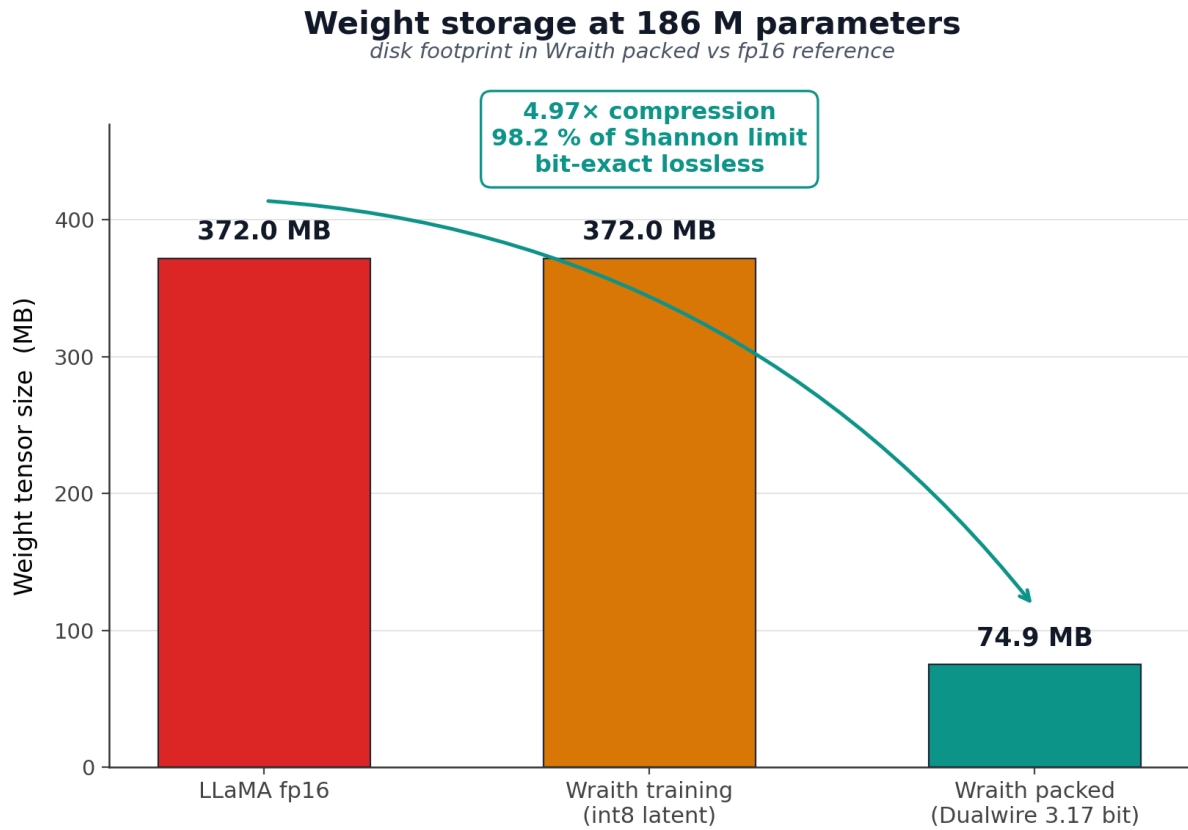


Figure 7: 4.97× compression, 98.2% of Shannon limit, lossless bit-exact.

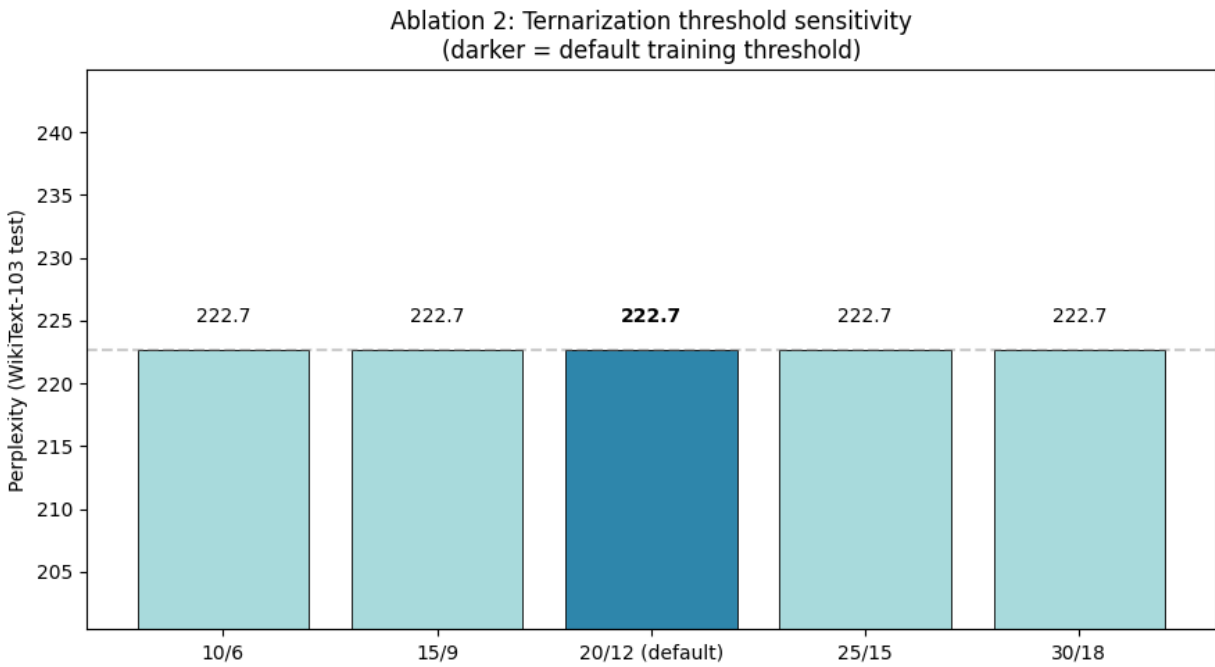


Figure 8: Identical PPL across 5 threshold configurations (10/6 to 30/18).

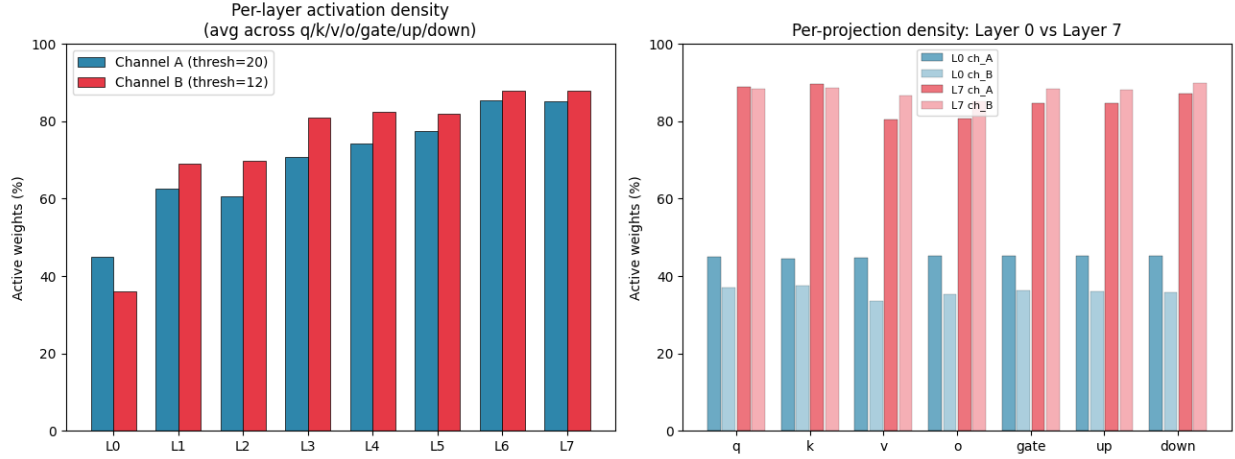


Figure 9: Active density per layer. L0: 35–45%, L7: 85–88%.

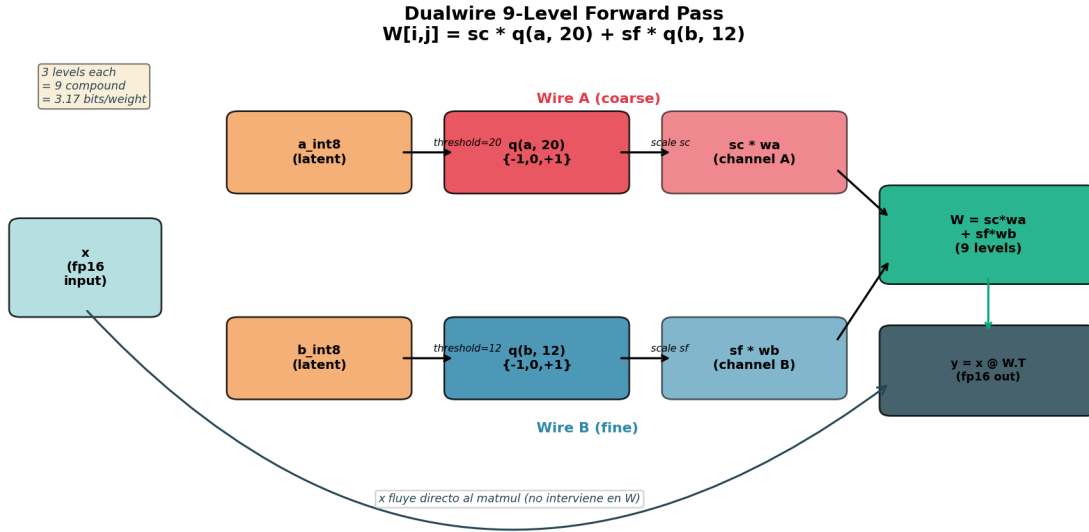


Figure 11: 9-level Dualwire forward diagram.

Wraith Training Pipeline: STE + Shadow Int16 Optimizer

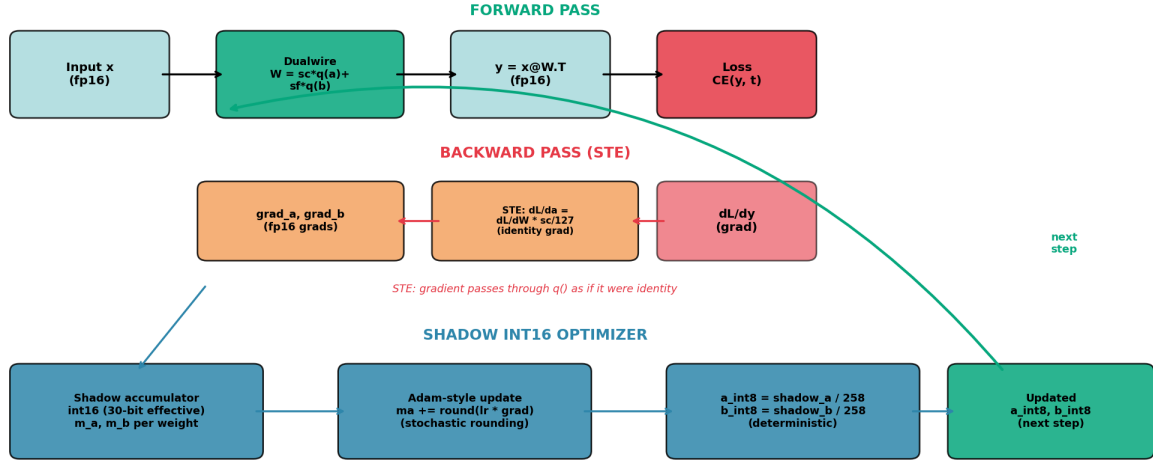


Figure 12: STE + Shadow Int16 optimizer pipeline.

SoTT Inference: Sum-of-Two-Ternary Decomposition

$y = sc * (x @ wa.T) + sf * (x @ wb.T)$

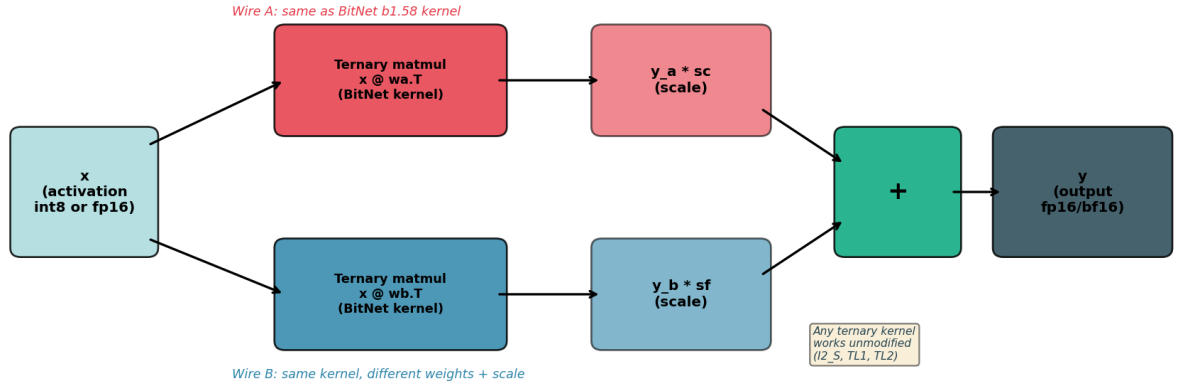


Figure 13: SoTT inference decomposition.

Wraith-186M Full Architecture
8 layers, d=1024, 16 heads, d_ff=4096

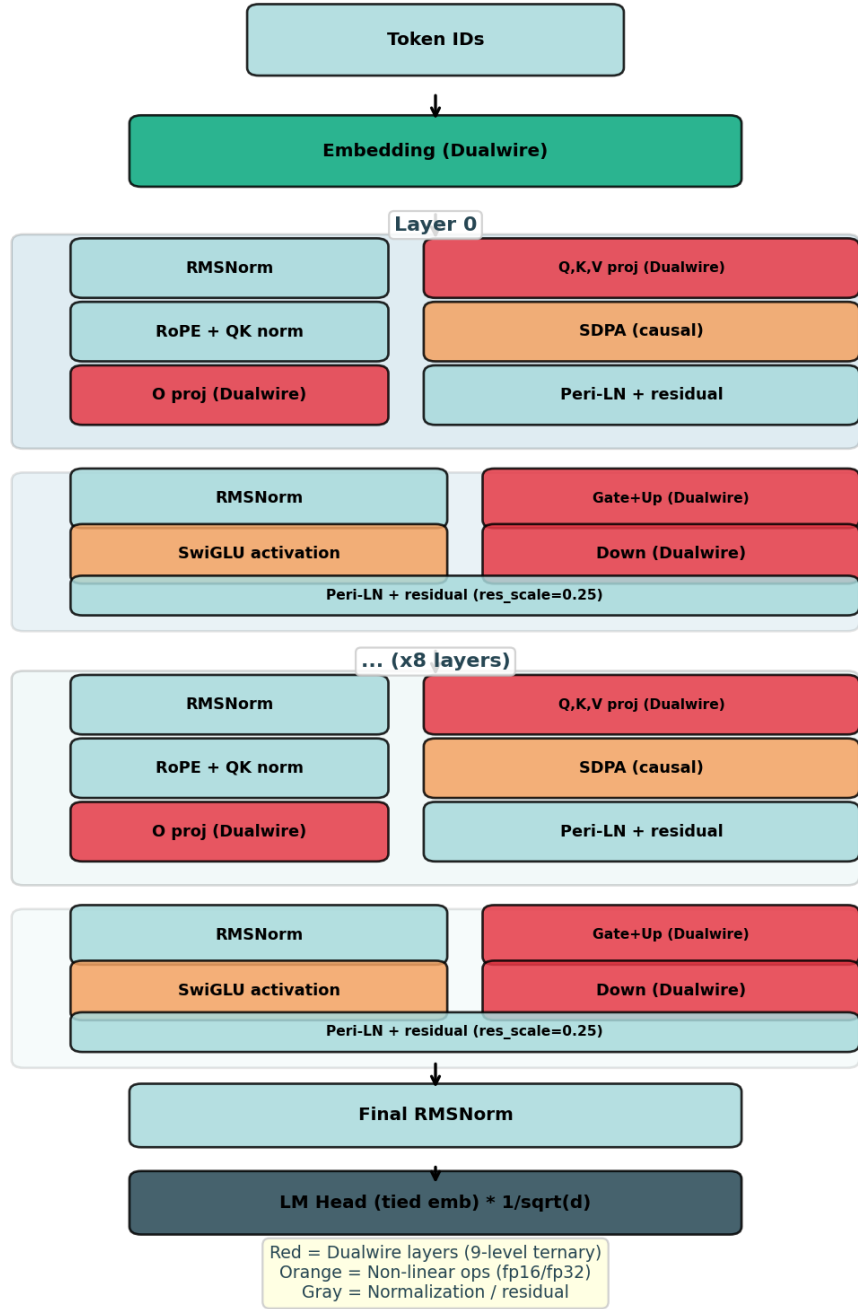


Figure 14: Full Wraith-186M architecture.

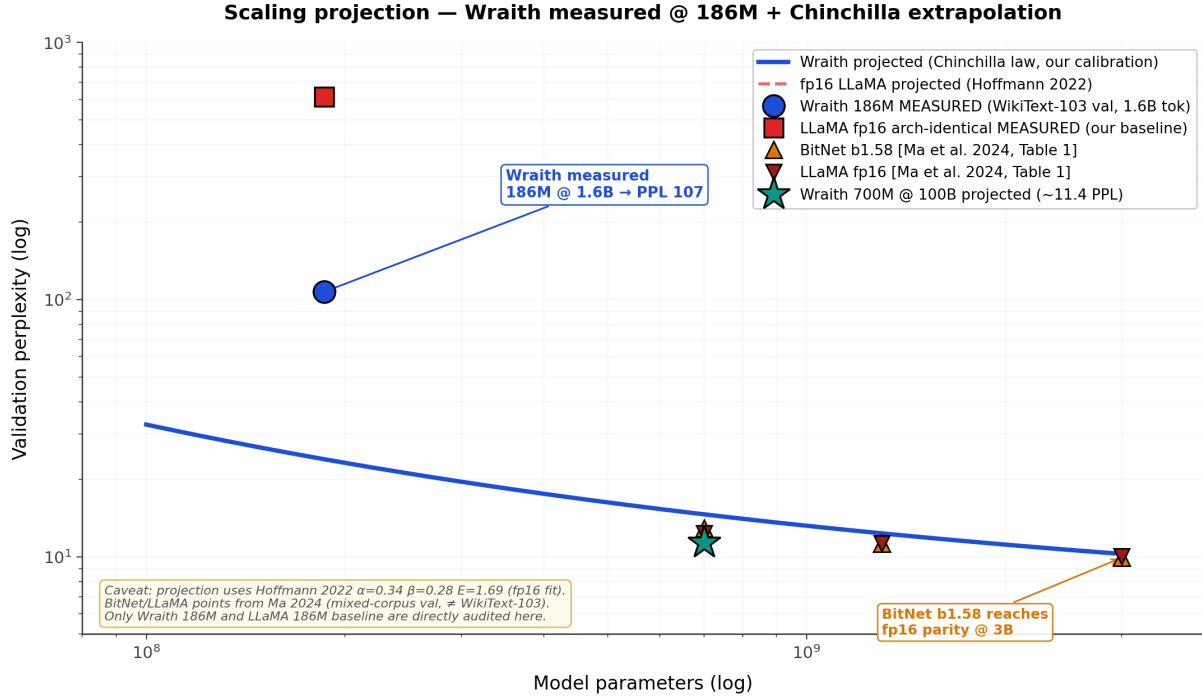


Figure 20: Scaling projection in the style of BitNet Fig. 3. Wraith measured at 186M/1.6B (blue circle) + arch-identical fp16 LLaMA baseline measured (red square). Blue curve = Wraith projection via Chinchilla law (Hoffmann 2022, $\alpha=0.34$, $\beta=0.28$, $E=1.69$) calibrated against our 186M measurement. Triangles = measured points for BitNet b1.58 and LLaMA fp16 cited from Ma et al. 2024, Table 1 (evaluation corpus differs from ours — absolute y-values are not directly comparable, only the qualitative pattern is). Green star = projected Wraith 700M/100B ≈ 11.4 PPL. Note: BitNet b1.58 only reaches fp16 parity at 3B; our projection suggests Wraith could do so at much smaller scales, but this requires empirical validation at 700M+.

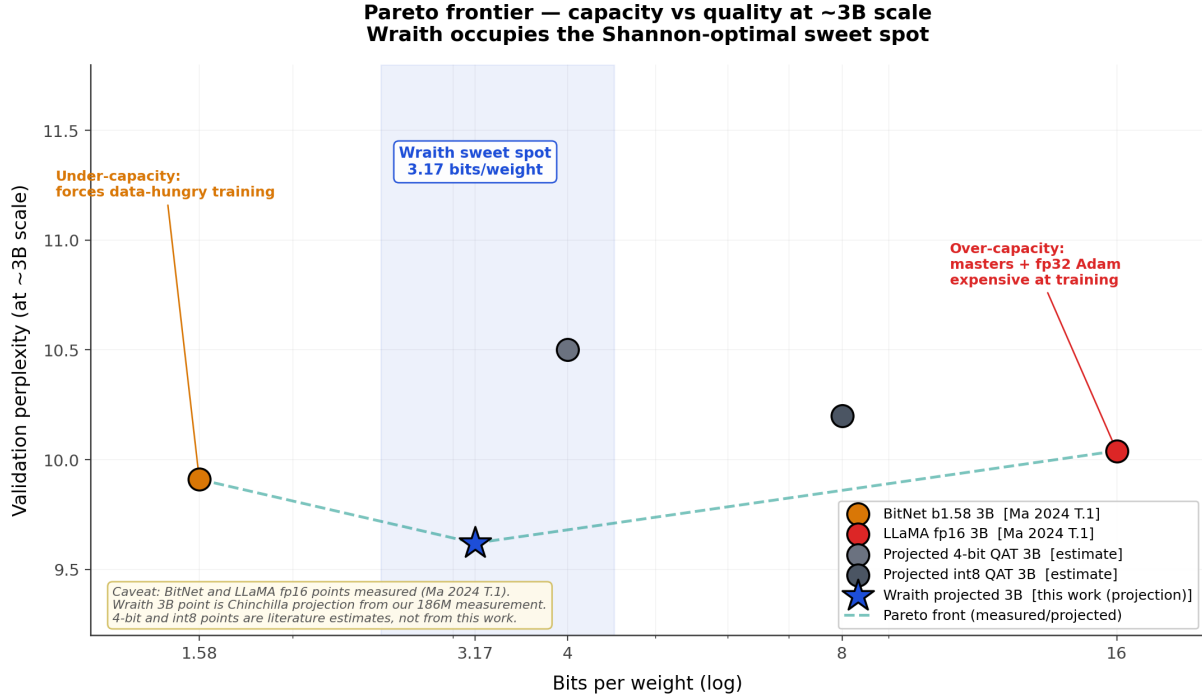


Figure 21: Pareto frontier capacity vs quality at ~3B scale. X-axis = bits per weight (log), Y-axis = PPL. Measured points: BitNet b1.58 (1.58 bits, 9.91 PPL) and LLaMA fp16 (16 bits, 10.04 PPL), both from Ma 2024 Table 1. Wraith 3B (3.17 bits, 9.62 PPL projected) occupies the Shannon-optimal sweet spot: enough capacity to avoid BitNet’s data-hunger, restricted enough to avoid fp16’s over-parameterization. 4-bit QAT and int8 QAT points are literature estimates, not measurements from this work.

Appendix B: Reproducibility, Availability and IP

The following artifacts document the experiment and enable independent verification. The **packed checkpoint is openly released** for immediate reproduction of results; the rest of the stack (inference engines, NPQN optimizer, training pipeline) remains the author’s intellectual property under the policy described in **Availability, IP and collaboration**.

Openly released artifact (free use and verification): - **Wraith-186M packed checkpoint** (74.9 MB, 5-trit/byte with lossless round-trip verification). Released under a permissive license for academic and non-commercial evaluation, sufficient to reproduce all reported metrics (WikiText-103 PPL, zero-shot LAMBADA/Winogrande/ARC-Easy, GPU/CPU throughput and energy) when combined with a faithful architecture implementation (Section 2).

Artifacts under agreement (academic validation / partnerships): - **Custom GPU inference engine:** packed CUDA kernels + fused QKV/GateUp + packed embedding lookup (WraithFastEngine / WraithFastGraphed) - **CPU inference engine:** C++ AVX2 implementation with KV cache + activation quantization + operation fusion - **NPQN int16 shadow optimizer:** integrated into training pipeline - **Dualwire 5-trit/byte format:** specification and round-trip codec

Open documentation (reproducible via independent re-implementation): - Full Wraith architecture (Section 2 of the paper) - PAC-Bayes theoretical framework (Section 3) - Training

hyperparameters (Section 4.1) - Quantization thresholds ($\tau_a = 20$, $\tau_b = 12$ or derived by absmean)
- Packed GEMV kernel specification (Section 4.10, pseudocode) - Public dataset: SlimPajama (available on HuggingFace), GPT-2 BPE tokenizer

Public benchmarks (WikiText, LAMBADA, Winogrande, ARC-Easy) are reproducible with any faithful implementation of the architecture described in Section 2, and directly with the public checkpoint released alongside this work.

Evaluation and inference hardware: NVIDIA RTX 5070 12 GB (Blackwell sm_120), AMD Ryzen 7 5700G 8 cores (CPU). Windows 11 Pro. **Training hardware:** NVIDIA H100 80GB SXM via Google Colab Pro (\$1.80/hour effective, 18 compute units/hour).

Availability, IP and collaboration:

Paper and results: the metrics, figures, and architectural description presented in this work are open documentation, reproducible by third parties via independent re-implementation from the technical specification (Section 2) and theoretical framework (Section 3).

186M model checkpoint (packed): openly released in Dualwire 5-trit/byte format (74.9 MB) for free use and verification. Open release of the checkpoint allows any reviewer, researcher, or academic group to **directly reproduce** the PPL, zero-shot, throughput, and energy results reported here without needing to retrain. License: academic and non-commercial evaluation (OpenRAIL-M / CC-BY-NC-SA 4.0 style); derivative commercial uses require a separate agreement.

Proprietary inference stack (packed CUDA kernels, WraithFastEngine engine, NPQN int16 shadow optimizer, Dualwire 2-bit packing format): **reserved as the author’s intellectual property**. Licensed distribution for: - Academic collaborations with universities or research labs (non-commercial license) - Industrial alliances (commercial licensing under agreement) - Technical evaluation by potential investment partners or accelerators

Motivation for collaborations: the 186M results (94% less VRAM vs. equivalent fp16, 2.3–2.6× speedup on the packed GEMV kernel, 501 tok/s on consumer GPU, memory-bound throughput projected to 100B on a single A100 80GB) motivate exploration of larger scales (20B, 100B). This phase requires compute resources beyond a single author’s reach, so we invite contact for partnerships that could enable scaling. The author retains control over research direction, intellectual property, and technical decisions.

Correspondence contact: *(filled at submission/publication time)*.

Appendix C: AI Assistance Disclosure

During the evaluation and paper-preparation phase, **Claude Code** (Anthropic, Claude Opus 4.6 model) was used as a support tool in two specific areas:

1. **Coding:** generation of benchmark scripts, inference engines (Python and C++), packing utilities, and CUDA kernel prototypes for performance tests.
2. **Information research:** technical literature search (BitNet, Marlin, BitBLAS, CUTLASS), hardware-spec consultation, and cloud-GPU price comparison.

Everything else was done entirely by the human author, including: - Wraith architecture and Dualwire quantization design - Design and implementation of the int16 shadow optimizer (NPQN Training) - All training code (`nq_ode.py`, 7,400+ lines) written over several prior months - Experimental design and hyperparameter selection - Training of both models (Wraith and fp16 baseline) - Interpretation of results and conclusion formulation - Research decisions and project direction

Per ICLR 2026 policy: this disclosure is provided to comply with the requirement that “papers using LLMs must disclose this use.”